All India Aakash Test Series for NEET - 2028

TEST - I (Code-I)

Click here for Code-J sol.

Test Date : 01/11/2026

ANSWERS

1.	(3)	37.	(2)	73.	(2)	109.	(1)	145.	(4)
2.	(2)	38.	(2)	74.	(4)	110.	(2)	146.	(1)
3.	(1)	39.	(4)	75.	(1)	111.	(3)	147.	(3)
4.	(4)	40.	(3)	76.	(1)	112.	(1)	148.	(4)
5.	(2)	41.	(1)	77.	(4)	113.	(3)	149.	(1)
6.	(2)	42.	(1)	78.	(3)	114.	(1)	150.	(2)
7.	(1)	43.	(2)	79.	(3)	115.	(2)	151.	(1)
8.	(2)	44.	(1)	80.	(2)	116.	(3)	152.	(3)
9.	(3)	45.	(3)	81.	(2)	117.	(3)	153.	(3)
10.	(4)	46.	(1)	82.	(4)	118.	(3)	154.	(2)
11.	(2)	47.	(1)	83.	(4)	119.	(1)	155.	(4)
12.	(2)	48.	(4)	84.	(1)	120.	(3)	156.	(3)
13.	(1)	49.	(2)	85.	(4)	121.	(3)	157.	(4)
14.	(1)	50.	(1)	86.	(1)	122.	(3)	158.	(1)
15.	(1)	51.	(3)	87.	(3)	123.	(1)	159.	(2)
16.	(2)	52.	(2)	88.	(1)	124.	(4)	160.	(2)
17.	(3)	53.	(3)	89.	(1)	125.	(2)	161.	(1)
18.	(4)	54.	(4)	90.	(4)	126.	(1)	162.	(4)
19.	(4)	55.	(3)	91.	(3)	127.	(2)	163.	(2)
20.	(3)	56.	(3)	92.	(1)	128.	(3)	164.	(1)
21.	(4)	57.	(3)	93.	(4)	129.	(4)	165.	(4)
22.	(2)	58.	(3)	94.	(3)	130.	(1)	166.	(3)
23.	(4)	59.	(4)	95.	(2)	131.	(3)	167.	(1)
24.	(3)	60.	(3)	96.	(3)	132.	(1)	168.	(2)
25.	(3)	61.	(4)	97.	(2)	133.	(2)	169.	(4)
26.	(3)	62.	(2)	98.	(2)	134.	(3)	170.	(3)
27.	(3)	63.	(1)	99.	(2)	135.	(4)	171.	(1)
28.	(1)	64.	(4)	100.	(3)	136.	(3)	172.	(4)
29.	(1)	65.	(3)	101.	(4)	137.	(3)	173.	(2)
30.	(2)	66.	(3)	102.	(2)	138.	(1)	174.	(1)
31.	(1)	67.	(2)	103.	(4)	139.	(2)	175.	(2)
32.	(1)	68.	(4)	104.	(1)	140.	(3)	176.	(4)
33.	(2)	69.	(1)	105.	(4)	141.	(1)	177.	(3)
34.	(3)	70.	(4)	106.	(3)	142.	(4)	178.	(4)
35.	(1)	71.	(2)	107.	(2)	143.	(2)	179.	(3)
36.	(2)	72.	(2)	108.	(2)	144.	(2)	180.	(4)

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HINTS & SOLUTIONS

[PHYSICS]

1. Answer (3)

Hint & Sol.: Least precise measurement may or may not be most accurate and vice-versa.

2. Answer (2)

Hint: Recall formula for surface area, $A = 6a^2$

Sol.: $A = 6a^2$ ...(1)

$$\Rightarrow \frac{\Delta A}{A} = \frac{2\Delta a}{a}$$

$= 2 \times 1\% = 2\%$

3. Answer (1)

Hint and Sol.: Using binomial expansion *i.e.*,

$(1+x)^n \approx 1+nx$ if $|x| << 1$

$$\therefore \left(1+\frac{h}{R}\right)^{-2} \approx 1-\frac{2h}{R}$$

4. Answer (4)

Hint: Check, if dimensions of LHS are equal to RHS, then equation is dimensionally correct otherwise incorrect.

Sol.: All first three options are standard results and they are dimensionally correct but option (4) is dimensionally incorrect.

since term $(u) \rightarrow$ speed $\rightarrow [LT^{-1}]$

and term $\left[\frac{1}{2}at^2\right] \rightarrow [LT^{-2}] \times [T^2] = [LT^0] = [L]$

and $[S] \rightarrow [L]$

So option (4) is dimensionally incorrect.

5. Answer (2)

Hint: Use principle of dimensional homogeneity and we know that exponent terms and their powers are dimensionless.

Sol.: $x = \frac{v_0}{\alpha} - \frac{v_0}{\alpha}e^{-\alpha t}$

$\Rightarrow \alpha t$ is dimensionless

$$[\alpha] = \left[\frac{1}{t}\right]$$

$$[\alpha] = [T^{-1}]$$

and $[x] = \left[\frac{v_0}{\alpha}\right]$

$$\Rightarrow [L] = \frac{[v_0]}{[T^{-1}]}$$

$$\Rightarrow [v_0] = [LT^{-1}]$$

$\therefore \alpha^2 v_0^3 \rightarrow [T^{-2}] \times [LT^{-1}]^3$

$= [L^3T^{-5}] = [M^0L^3T^{-5}]$

6. Answer (2)

Hint and Sol.: Velocity = $\frac{\text{Displacement}}{\text{time}}$

$= \frac{\text{Length}}{\text{Time}}$

$\therefore$ Velocity is derived quantity

7. Answer (1)

Hint and Sol.:

$$\text{Power} = \frac{\text{Work}}{\text{Time}} \rightarrow \frac{\text{Joule}}{\text{Second}} \rightarrow \text{Watt}$$

So, kilowatt $\rightarrow 10^3$ watt

8. Answer (2)

Hint: Density, $\rho = \frac{m}{V} = \frac{m}{\pi r^2 l}$

Sol.: $\Rightarrow \left(\frac{\Delta\rho}{\rho}\right)_{\text{maximum}} = \left(\frac{\Delta m}{m} + \frac{2\Delta r}{r} + \frac{\Delta l}{l}\right)$

$$\left(\frac{\Delta\rho}{\rho}\right)_{\text{max}} = \frac{0.003}{0.60} + \frac{2\times 0.01}{0.50} + \frac{0.05}{10.00}$$

$$= \frac{0.1}{20} + \frac{2}{50} + \frac{5}{1000}$$

$$= \frac{5+40+5}{1000} = \frac{50}{1000} = \frac{5}{100}$$

Maximum percentage error in density = 5%

9. Answer (3)

Hint: Use principle of dimensional homogeneity i.e., each term has same dimensions.

Sol.: $[A] = [L]$

$[B] = [L]$

$[C] = [LT^{-2}]$

and $[D] = [L]$

$$\therefore \left[\frac{ABC}{D}\right] = \left[\frac{L \times L \times LT^{-2}}{L}\right] = [L^2T^{-2}]$$

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10. Answer (4)

Hint: Area under this *v*-*t* graph gives distance covered.

Sol.: Area bounded in $t = 30.5$ second

$$= \frac{1}{2}(400+200)\times 2 + 400\times(30.5-2)$$

$= 300 \times 2 + 400 \times 28.5$

$= 600 + 11400$

$= 12000$ m $= 12$ km

11. Answer (2)

Hint: In multiplication/division, final result will have significant figures as equal to any data which has least significant figure.

Sol.: Here least significant figures are three so simply by observation we can mark option (2) because $x = 1330$ has 3 significant figure.

Note: You can also do mathematical operation, and then finally round-off the result upto 3 significant figure.

12. Answer (2)

Hint: $LC = \dfrac{\text{Pitch}(x)}{\text{Number of divisions on circular scale } (n)}$

Sol.: $LC = \dfrac{x}{n}$

Initially $0.01 = \dfrac{x}{n}$

finally, $LC' = \dfrac{\left(x + \dfrac{75}{100}x\right)}{n - \dfrac{n\times 50}{100}}$

$$LC' = \frac{\frac{175}{100}x}{\frac{n}{2}}$$

$$LC' = \frac{350}{100}\times\frac{x}{n} = 3.5\times 0.01 = 0.035 \text{ mm}$$

13. Answer (1)

Hint and Sol.:

(A-Q) $v = u - gt = -gt + u$

Graph will be straight line with negative slope

(B-R) $|v| = |u - gt|$

Graph will be always positive and straight line

(C-P) $h = ut - \dfrac{1}{2}gt^2$ parabola with downward opening.

14. Answer (1)

Hint: Use velocity $v = \dfrac{dx}{dt}$

and acceleration $a = \dfrac{dv}{dt}$

Sol.: $x = (2t+5)^2$

$\Rightarrow x = 4t^2 + 20t + 25$

$\Rightarrow v = \dfrac{dx}{dt} = 8t + 20$

$\Rightarrow a = \dfrac{dv}{dt} = 8$ = constant

15. Answer (1)

Hint and Sol.: In case of uniform motion, speed and velocity are constant. Therefore, all statements A, B and C are correct

16. Answer (2)

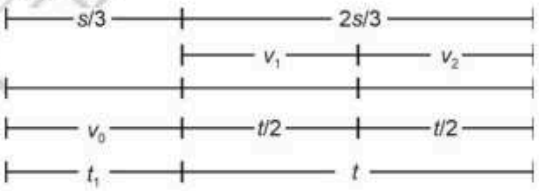
Hint: Use $s_{n^{th}} = u + \dfrac{a}{2}[2n-1]$

Sol.: $\dfrac{s_{3rd}}{s_{5th}} = \dfrac{S_1}{S_2} = \dfrac{\frac{a}{2}[2\times 3-1]}{\frac{a}{2}[2\times 5-1]} = \dfrac{5}{9}$

17. Answer (3)

Hint: Average speed $= \dfrac{\text{Total distance covered}}{\text{Time taken}}$

Sol.:



here, $t_1 = \dfrac{s}{3v_0}$, $v_1\dfrac{t}{2} + \dfrac{v_2}{2}t = 2\dfrac{s}{3}$

$\Rightarrow t = \dfrac{4s}{3(v_1+v_2)}$

Now, Average speed $= \dfrac{s}{t_1 + t}$

$$= \frac{s}{\frac{s}{3v_0} + \frac{4s}{3(v_1+v_2)}}$$

$$= \frac{1}{\left[(v_1+v_2)+4v_0\right]/3v_0(v_1+v_2)}$$

$$= \frac{3v_0(v_1+v_2)}{(v_1+v_2)+4v_0}$$

18. Answer (4)

Hint and Sol.: For coming to rest, acceleration and velocity must be in opposite direction. So, angle should be 180° (π radian)

19. Answer (4)

Hint: Use the concept of relative velocity and equations of uniformly accelerated motion.

Sol.: Stopping distance $= \frac{u^2}{2a} = \frac{(4)^2}{2\times 1.2} = 6.66$ m < 10 m

$\Rightarrow$ He will never be able to catch bus.

20. Answer (3)

Hint and Sol.:

Acceleration, $a = \frac{dv}{dt} \Rightarrow \int dv = \int a dt$

$\Rightarrow \Delta v = \int a dt$ = area under acceleration vs time graph

21. Answer (4)

Hint: Use $v^2 = u^2 + 2as$

Sol.: $v^2 = u^2 + 2as$

$0 = u^2 - 2as$

$s = \frac{u^2}{2a}$

here $a \rightarrow$ constant

$\therefore \ s \propto u^2$

when u is made n times, then

s will become n^2 times.

22. Answer (2)

Hint: Here acceleration and velocity are in opposite direction so particle will return back on same path. We have to check reversal instant (when $v = 0$) is in between given time interval or outside the given interval. Then we decide whether, distance is equal to displacement or not; if not, then distance = actual path length

Sol.: At reversal point $v = 0$

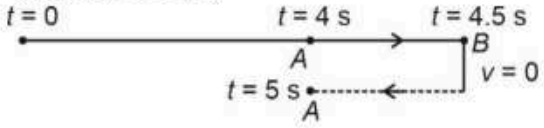
then time at which it occurs

$v = u + at$

$\Rightarrow 0 = 9 - 2t \qquad \Rightarrow t = 4.5$ second

so this is in between $t = 4$ s to $t = 5$ s

(i.e., fifth second)



here required distance = $|AB| + |BA|$

$= 2|BA|$ ($\because$ acceleration $\rightarrow$ constant)

$= 2\times\frac{1}{2}\times 2\times\left(\frac{1}{2}\right)^2$

$= 2\times\frac{1}{4} = \frac{1}{2} = 0.5$ m

23. Answer (4)

Hint and Sol.: Acceleration, $a = \frac{dv}{dt}$

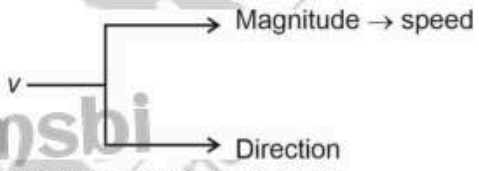
from this, (i) if $v \rightarrow$ constant

then $a = 0$ at all time

(ii) If v is continuously varying then $\frac{dv}{dt} \neq 0$, even $v = 0$ at any instant (in case of reversal point of motion)

24. Answer (3)

Hint and Sol.: $a = \frac{dv}{dt}$; and



From here we can conclude

If velocity is changing, we cannot surely say whether speed is changing or not as well as whether direction is changing or not.

25. Answer (3)

Hint: $v_{PQ} = v_P - v_Q$ and slope of v-t graph gives acceleration

Sol.: $v_P = a_P t$ and

$v_Q = a_Q t$

$\therefore \ v_{PQ} = v_P - v_Q = (a_P - a_Q)t$

here $a_P - a_Q$ is positive constant

$\Rightarrow v_{PQ} = Kt$

i.e., relative velocity will increase linearly with time.

26. Answer (3)

Hint: Instantaneous acceleration, $a = \frac{dv}{dt}$

Sol.: $A = \frac{dv}{dt} \rightarrow dv = A \cdot dt$

$\Rightarrow \int dv = \int A dt$

@shivamshi

$$\Rightarrow \int_u^v dv = \int_0^t at \cdot dt = \frac{at^2}{2}$$

$$\Rightarrow v - u = \frac{at^2}{2} \Rightarrow v = u + \frac{at^2}{2}$$

27. Answer (3)

Hint: Use, $h = ut - \frac{1}{2}gt^2$

Sol.: Using $h = ut - \frac{1}{2}gt^2$

According to question

$h_1 = h_2$

$$\Rightarrow ut - \frac{1}{2}gt^2 = u(t-5) - \frac{1}{2}g(t-5)^2$$

$$\Rightarrow ut - 5t^2 = u(t-5) - 5(t-5)^2$$

$$\Rightarrow -5t^2 = -5u - 5(t-5)^2$$

$$\Rightarrow -t^2 = -u - (t-5)^2$$

$$\Rightarrow t^2 - (t-5)^2 = 40 \quad [\because u = 40 \text{ m/s}]$$

$$\Rightarrow t^2 - t^2 + 10t - 25 = 40$$

$10t = 65 \Rightarrow t = 6.5$ second

Now, $h = ut - \frac{1}{2}gt^2$

$$h = 40 \times 6.5 - \frac{1}{2}10 \times (6.5)^2$$

$= 260 - 5 \times 6.5 \times 6.5$

$= 260 - 211.25 = 48.75$ metre

28. Answer (1)

Hint: Slope of position(x)-time(t) graph gives velocity.

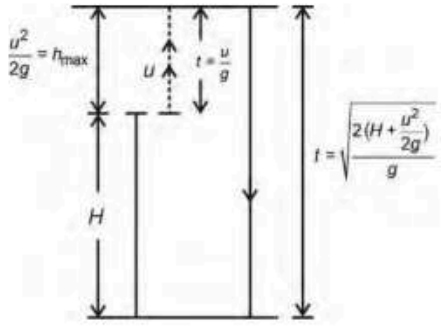
Sol.: From graph it is clear that at 6 points slope is zero *i.e.*, $v = 0$, so particle comes to rest 6 times.

29. Answer (1)

Hint: Maximum height attained by stone (dropped from B_1) is least, so total time of journey for stone dropped from B_1 is least.

Sol.: Total journey time for a stone is given by

$$T = \frac{u}{g} + \sqrt{\frac{2\left(H + \frac{u^2}{2g}\right)}{g}}$$



From above expression of T, it is clear that

If u is higher, then T will be higher

30. Answer (2)

Hint: For uniform acceleration

$v_{av} = \frac{u+v}{2}$ and

$s = v_{av} \times$ time

Sol.: $v = \frac{0+144}{2} = 72$ km/h

$$= \frac{72 \times 5}{18} = 20 \text{ m/s}$$

Now $s = v_{av} \times t = 20 \times 20 = 400$ m

31. Answer (1)

Hint and Sol.: Distance covered is always increasing for a moving body.

Therefore speed, $\frac{ds}{dt} \rightarrow$ positive

32. Answer (1)

Hint: Time taken to cross each other

$$= \frac{\text{Sum of total length of trains}}{\text{relative speed of trains}}$$

Sol.: Crossing time $= \frac{100+60}{|v_A| + |v_B|}$

$$4 = \frac{160}{v + 3v} = \frac{160}{4v}$$

$$\Rightarrow v = \frac{160}{16} = 10 \text{ m/s}$$

so $v_A = 10$ m/s and $v_B = 30$ m/s

33. Answer (2)

Hint and Sol.: s always denotes displacement covered while u and a may be positive, or negative or zero.

34. Answer (3)

Hint and Sol.: $a = \frac{dv}{dt}$ *i.e.*, slope of *v-t* graph gives acceleration.

For first half-time slope is negative and constant while for second half-time slope is positive and constant, so option (3) is correct.

35. Answer (1)

Hint: Learn it, as a standard case or use $h = ut - \frac{1}{2}gt^2$ and solve for its roots t_1 and t_2

Sol.: Here $h = ut - \frac{1}{2}gt^2$

It has two values (roots) t_1 and t_2 for same *h*.

so, converting this equation in standard quadratic form

$2h = 2ut - gt^2$

$\Rightarrow gt^2 - 2ut + 2h = 0$

Now using condition for products of roots for $ax^2 + bx + c = 0$

$\Rightarrow \alpha\beta = \frac{c}{a}$

$\Rightarrow t_1 t_2 = \frac{2h}{g}$

$\Rightarrow h = \frac{1}{2}gt_1t_2$

36. Answer (2)

Hint: Use, $v = u + at$

Sol.: $-3u = u - gt$

$\Rightarrow t = \frac{4u}{g}$

37. Answer (2)

Hint: Use $v = \frac{dx}{dt}$ and $a = \frac{dv}{dt}$

Sol.: $x = \frac{1}{2}t^2 - \frac{1}{3}t^3$

$v = \frac{dx}{dt} = \frac{d}{dt}\left[\frac{1}{2}t^2 - \frac{1}{3}t^3\right]$

$= \frac{1}{2} \times 2t - \frac{1}{3} \times 3t^2$

$v = t - t^2$

and $a = \frac{dv}{dt} = \frac{d}{dt}(t - t^2)$

$a = 1 - 2t$

here $a = 0$

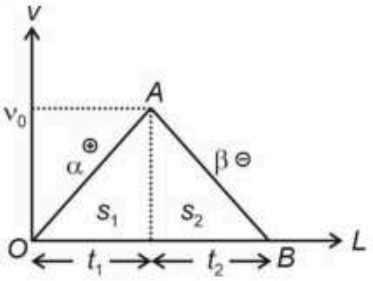
$\Rightarrow 1 - 2t = 0$

$t = \frac{1}{2}$ second

38. Answer (2)

Hint: Draw *v-t* graph and use $v = u + at$ as well as $s = \frac{1}{2}at^2$

Sol.:



For motion *O* to *A*:

$v_0 = \alpha t_1 \Rightarrow t_1 = \frac{v_0}{\alpha}$ ...(i)

For motion *A* to *B*:

$0 = v_0 - \beta t_2$

$\Rightarrow t_2 = \frac{v_0}{\beta}$ ...(ii)

$\Rightarrow \frac{t_1}{t_2} = \frac{\beta}{\alpha}$

Now area of *v-t* graph gives displacement

$\Rightarrow$ *O* to *A*:

$s_1 = \frac{1}{2}v_0t_1$ ...(iii)

and *A* to *B*:

$s_2 = \frac{1}{2}v_0t_2$ ...(iv)

$\Rightarrow \frac{s_1}{s_2} = \frac{t_1}{t_2}$

39. Answer (4)

Hint: Focus on definition of physical quantity and write expression

$s = \frac{E}{A \times t}$

Sol.: $[s] = \frac{[ML^2T^{-2}]}{[L^2 \times T]} = [ML^0T^{-3}]$

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40. Answer (3)

Hint: There will be error of $\frac{1}{5}$ second in measured time t = 15 second and here $t = nT$

Sol.: $t = nT$

$\Rightarrow \Delta t = n\Delta T$

$\Rightarrow \frac{\Delta t}{t} = \frac{\Delta T}{T}$

$\therefore \frac{\Delta t}{t} = \frac{\frac{1}{5}}{15} = \frac{1}{75}$

Percentage error in $(T) = \frac{100}{75} = \frac{4}{3} = 1.3\%$

41. Answer (1)

Hint and Sol.: Using rules for counting significant figures, 0.01 has one significant figure and 0.50 has two significant figure and (R) explains (A).

42. Answer (1)

Hint and Sol.: Here digits to be dropped are 6 and 7, so preceding digit is increased by one.

43. Answer (2)

Hint: Errors are very small, so, differentiate the equation and find $\frac{dy}{y}$ to find fractional error in y

Sol.: $y = k\tan x$

$\Rightarrow dy = k\sec^2 x dx$

$\Rightarrow \frac{dy}{y} = \frac{\sec^2 x dx}{\tan x} = \frac{dx}{\cos^2 x \times \frac{\sin x}{\cos x}}$

$= \frac{dx}{\sin x \cos x} = \frac{2dx}{\sin 2x}$

44. Answer (1)

Hint and Sol.: Only multiplication or division is possible for quantities having different dimensions.

45. Answer (3)

Hint: Write G in terms of c, g and p as $[G] = [c^a g^b p^c]$ and now write dimension of both sides of equation and equate dimension of M, L and T, to find a, b and c.

Sol.: $[G] = [LT^{-1}]^a [LT^{-2}]^b \cdot [ML^{-1}T^{-2}]^c$

$\Rightarrow \left[\frac{MLT^{-2} \times L^2}{M \cdot M}\right] = [M^c L^{a+b-c} T^{-a-2b-2c}]$

$\Rightarrow [M^{-1}L^3T^{-2}] = [M^c L^{a+b-c} T^{-a-2b-2c}]$

$\Rightarrow c = -1,\ a + b - c = 3$ and $-a - 2b - 2c = -2$

from these equations we have

$c = -1,\ a + b = 2$ ...(i)

and $a + 2b = 4$ ...(ii)

After solving (i) and (ii), we get

$c = -1,\ a = 0$ and $b = 2$

so, $[G] = [c^0 g^2 p^{-1}]$

[CHEMISTRY]

46. Answer (1)

Hint: Homogeneous mixture has uniform composition throughout the mixture

Sol.: Aqueous NaCl and air are homogeneous mixture.

47. Answer (1)

Hint: Kilogram is SI unit of mass.

Sol.: Mole $\Rightarrow$ Amount of substance

Candela $\Rightarrow$ Luminous intensity

Ampere $\Rightarrow$ Electric current

48. Answer (4)

Hint: Number of molecule = mole N_A

Sol.: Initial molecules $= \frac{2.24 \times 10^{-3}}{22.4} \times N_A$

$= 10^{-4}\ N_A$

Removed molecules $= \frac{2.1 \times 10^{-3}}{28} \times N_A$

$= 0.75 \times 10^{-4}\ N_A$

Left molecule $= 10^{-4}\ N_A - 0.75 \times 10^{-4}\ N_A$

$= 0.25 \times 10^{-4}\ N_A$

$= 2.5 \times 10^{-5} \times 6.02 \times 10^{23}$

$= 1.5 \times 10^{19}$

49. Answer (2)

Hint: $2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$

$2KCl + H_2SO_4 \rightarrow K_2SO_4 + 2HCl$

Sol.: 117 g NaCl gives 142 g Na_2SO_4

hence x g NaCl gives $\frac{142}{117} \times$ 'x' g Na_2SO_4

(1.8062 – x) g KCl will give

$\frac{174}{149}(1.8062 - x)$ g K_2SO_4

$\frac{142x}{117} + \frac{174(1.8062 - x)}{149} = 2.1568$

21158x + 36770.62 – 20358x = 37599.49

800x = 828.87

x = 1.03 g

1.8062 – x = 0.7701 g

$\%NaCl = \frac{1.03}{1.8062} \times 100$

= 57.36%

%KCl = 42.64%

50. Answer (1)

Hint: On heating, water of hydration will be removed.

Sol.: $Na_2SO_4 \cdot xH_2O \xrightarrow{\Delta} Na_2SO_4 + xH_2O$

Mole(s) of anhydrous $NaSO_4 = \frac{\text{mole of } H_2O}{x}$

$\frac{44.1}{142} = \frac{55.9}{18 \times x}$

x = 10

51. Answer (3)

Hint: Only acidic gas gets absorbed in basic KOH pellets

Sol.:

	$HCOOH \xrightarrow{\text{conc. } H_2SO_4}$	$H_2O(l)$ +	$CO(g)$
t = 0	0.1 mol	0	0
t = t	0	0.1	0.1

	$H_2C_2O_4 \xrightarrow{\text{conc. } H_2SO_4}$	$H_2O(l)$ +	$CO(g)$ +	$CO_2(g)$
t = 0	0.1 mol	0	0	0
t = t	0	0.1	0.1	0.1

Since CO_2 will get absorbed in KOH, remaining gaseous product is CO gas only

mole of CO gas = 0.2

mass of CO gas = 0.2 × 28

= 5.6 g

52. Answer (2)

Hint: $N_2 + 3H_2 \rightarrow 2NH_3$

H_2 will be limiting reagent

Sol.:

	N_2	+ $3H_2$	$\rightarrow 2NH_3$
t = 0	14 g	2 g	
	0.5 mol	1 mol	0
t = t	0.5 – 0.33	0	0.66 mol

mass of NH_3 formed $= \frac{2}{3} \times 17 = 11.33$ g

volume of NH_3 produced at STP $= \frac{2}{3} \times 22.4$

= 14.93 L

53. Answer (3)

Hint: Empirical formula represents simplest ratio of number of atoms of constituents present in one molecule.

Sol.: For benzene molecular formula is C_6H_6 and empirical formula is CH and for acetylene molecular formula is C_2H_2 and empirical formula is CH.

54. Answer (4)

Hint: 1000 g of solvent contains 180 g of urea

Sol.: Mass of solution = (1000 + 180) g = 1180 g

1.18 kg of solution contains 1000 g of water

1.8 kg of solution contains $\frac{1000 \times 1.8}{1.18}$ g of water

Mass of water = 1.5 kg

55. Answer (3)

Hint: According to Dalton atomic theory the matter is composed of extremely small indivisible particles.

Sol.: Atoms of different element combine with each other in a fixed whole number ratio to form compounds

56. Answer (3)

Hint: Mole of $HNO_3 = 700 \times 3 \times 10^{-3} = 2.1$

Sol.: Mass of $HNO_3 = 2.1 \times 63 = 132.3$ g

70 g HNO_3 is present in 100 g of solution

132.3 g HNO_3 is present in $\frac{100 \times 132.3}{70} = 189$ g

57. Answer (3)

Hint: 6.00 has three significant figure while 600 has only one.

Sol.: Mass percentage of oxygen in glucose $= \frac{16 \times 6 \times 100}{180} = 53.3\%$.

58. Answer (3)

Hint: Volume of one mole gas at STP is 22.4 L

Sol.:

$d = \frac{\text{mass}}{\text{volume}}$

$= \frac{28}{22.4}$ = 1.25 g/L

59. Answer (4)

Hint: For mixing

$$[H^+] = \frac{N_1V_1 + N_2V_2 + N_3V_3}{V_1 + V_2 + V_3}$$

Sol.: $[H^+] = \frac{200\times0.4 + 300\times0.1 + 500\times0.4}{200 + 300 + 500}$

$= \frac{80 + 30 + 200}{1000}$ = 0.31 M

60. Answer (3)

Hint: Mole of 0.15 g $CO_3^{2-} = \frac{0.15}{60} = 2.5 \times 10^{-3}$

Sol.: Number of ions = $2.5 \times 10^{-3} \times 6.02 \times 10^{23}$

$= 1.5 \times 10^{21}$ ions

Number of O atoms = $3 \times 1.5 \times 10^{21} = 4.5 \times 10^{21}$

mass of carbon = $0.0025 \times 12 = 0.03$ g

61. Answer (4)

Hint: For minimum molecular mass, given molecule must have 1, Mg atom

Sol.: $\%Mg = \frac{1\times24}{M}\times100$

$0.01 = \frac{24}{M}\times100$

$M = 2.4 \times 10^5$ u

62. Answer (2)

Hint: % H = 100 – (%C + %O)

= 100 – (54.54 + 36.37)

= 9.09%

Sol.:

	C	H	O
Mass% :	54.54	9.09	36.37
Mole ratio :	54.54	9.09	36.37
	12	1	16
	4.54	9.09	2.27
Simplest ratio :	2	4	1

Empirical formula C_2H_4O

Empirical formula mass = 44 u

63. Answer (1)

Hint: $C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$

Sol.: Number of mole of $C_2H_6 = \frac{3}{30} = 0.1$

1 mole C_2H_6 requires 3.5 mol O_2

0.1 mole C_2H_6 requires 0.35 mol O_2 = 11.2 g

Volume of CO_2 obtained at STP = $0.1 \times 2 \times 22.4$

= 4.48 L

64. Answer (4)

Hint: S is limiting reagent for given reaction

Sol.:

$Mg(s) + S(s) \rightarrow MgS(s)$

$\frac{4}{24}$ $\frac{4}{32}$

$\frac{1}{6}$ mol $\frac{1}{8}$ mol

mole of MgS = $\frac{1}{8}$

mass of MgS = $\frac{1}{8}\times56 = 7$ g

65. Answer (3)

Hint: $°F = \left(\frac{9}{5}\right)\times°C + 32$

Sol.: For 25°C,

$F = \frac{9}{5}\times25 + 32 = 77°F$

For 40°C,

$F = \frac{9}{5}\times40 + 32 = 104°F$

For –60°C,

$F = \frac{9}{5}\times(-60) + 32 = -76°F$

For 100°C,

$F = \frac{9}{5}\times100 + 32 = 200°F$

66. Answer (3)

Hint: $E_{metal\ oxide} = E_m + E_0$

Sol.: $\frac{E_m}{E_0} = \frac{60}{40}$

$E_m = \frac{3}{2}\times8 = 12$

$E_{metal\ oxide}$ = 12 + 8 = 20

67. Answer (2)

Hint: For neutralisation number of gram equivalents of acid must be equal to number of gram equivalents of base.

Sol.: $(n_{eq})_{acid} = 100\times0.1 + 100\times0.5 = 60$

$(n_{eq})_{NaOH} = 60$

$0.2 \times V = 60$

V = 300 mL

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68. Answer (4)

Hint: According to JJ Thomson atom is uniform sphere of positive charge in which negative charge is embedded into it in such a manner as to give the most stable electrostatic arrangement.

Sol.: Presence of nucleus in an atom is confirmed by Rutherford's α-particle experiment.

$$\frac{r_{nucleus}}{r_{atom}} = 10^5$$

$$\frac{m_{proton}}{m_{electron}} = \frac{1.67\times10^{-27}\ kg}{9.109\times10^{-31}\ kg} = 1837$$

69. Answer (1)

Hint: Isobars have same mass number but different atomic numbers

Sol.: $A_1 = Z_A + n_A$

$100 = 48 + n_A$

$n_A = 52$

$A_2 = Z_B + n_B$

$100 = 52 + n_B$

$n_B = 48$

70. Answer (4)

Hint: Avg atomic mass $\frac{\sum A_i X_i}{100}$

$A_i \Rightarrow$ mass number of ith isotope

$X_i \Rightarrow$ percentage abundance of

Sol.: % abundance of heavier isotope = x

% abundance of lighter isotope = 100 – x

$$\frac{x\times22+(100-x)20}{100} = 20.3$$

$22x + 2000 - 20x = 2030$

$2x = 30$

$x = 15\%$

71. Answer (2)

Hint: For Mn atom last electron will be filled in 3*d* orbital.

Sol.:

$Mg \Rightarrow n = 3, l = 0, m = 0, s = +\frac{1}{2}$

$K \Rightarrow n = 4, l = 0, m = 0, s = +\frac{1}{2}$

$Br \Rightarrow n = 4, l = 1, m = -1, s = +\frac{1}{2}$

$Mn \Rightarrow n = 3, l = 2, m = +1, s = +\frac{1}{2}$

72. Answer (2)

Hint: For microscopic species

$$\Delta x\Delta p \geq \frac{h}{4\pi}$$

Sol.: $\Delta x\Delta v = \frac{h}{4\pi m}$

$$(\Delta x)^2 = \frac{h}{4\pi m}$$

$$\Delta x = \left(\frac{h}{4\pi m}\right)^{1/2}$$

$$\sqrt{\frac{h}{4\pi m}}\times\Delta p = \frac{h}{4\pi}$$

$$\Delta p = \sqrt{\frac{hm}{4\pi}}$$

73. Answer (2)

Hint: de-Broglie wavelength

$$\lambda = \frac{h}{\sqrt{2mKE}}$$

Sol.: $\lambda = \frac{6.6\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\times28\times10^{-23}}}$

$= \frac{6.6\times10^{-34}}{7.13\times10^{-27}} = 0.93 \times 10^{-7}$ m

74. Answer (4)

Hint: 'n' is a positive integer which is always greater than 'm'

Sol.: m, varies from '+l' to '–l'

s can be only $\frac{+1}{2}$ or $\frac{-1}{2}$

75. Answer (1)

Hint: Number of spherical node = n – l – 1

Sol.: For 5*p* orbital

Number of spherical mole = 5 – 1 – 1 = 3

d_{xy} orbital has two nodal planes.

76. Answer (1)

Hint: Number of orbital in a shell = n^2

Sol.: Number of orbital in 4th shell = $4^2 = 16$

As atomic number increases energy of orbital decreases.

For monoelectronic species energy depends on principal quantum number only.

Splitting of spectral line in magnetic field is Zeeman effect.

77. Answer (4)

Hint: $Cu \Rightarrow [Ar]3d^{10}4s^1$

Sol.: Number of electron with $m_l = 0$ are 13

78. Answer (3)

Hint: The element with atomic number 24 is chromium.

Sol.: Cr: $3d^5 4s^1$

$\therefore$ 6 unpaired electrons are present.

79. Answer (3)

Hint: Electron has maximum e/m ratio.

Sol.: Microwave has higher frequency than radio wave.

80. Answer (2)

Hint: For a black-body radiation, wavelength of intensity maxima decreases with increasing temperature.

Sol.: Since $\lambda_3 < \lambda_2 < \lambda_1$

hence $T_3 > T_2 > T_1$

81. Answer (2)

Hint: KE = E_{photon} – Work function

Sol.: KE = $6.6 \times 10^{-34} \times 10^{15} - 3.44 \times 10^{-19}$

$= 3.16 \times 10^{-19}$ J

82. Answer (4)

Hint: For single electron species

$$E_n = \frac{-13.6Z^2}{n^2}$$

Sol.: $$\frac{(E_3)_{Be^{3+}}}{(E_3)_{Li^{2+}}} = \frac{\frac{-13.6\times(4)^2}{(3)^2}}{\frac{-13.6\times(3)^2}{(3^2)}}$$

$$\frac{(E_3)_{Be^{3+}}}{x} = \frac{16}{9}$$

$$(E_3)_{Be^{3+}} = \frac{16x}{9}$$

83. Answer (4)

Hint: For single electron energy of orbital depends only on principal quantum number.

Sol.: H^- is two electron species

Hence 2nd excited state will be 2*p* subshell where value of *n* is 2.

84. Answer (1)

Hint: $\bar{\nu} = R_H \times Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

Sol.: $n_2 = 5$, $n_1 = 2$

$$\bar{\nu} = R_H \times (2)^2\left(\frac{1}{4} - \frac{1}{25}\right) = R_H\frac{21}{25}$$

85. Answer (4)

Hint: Mole of glucose = $200 \times 0.25 \times 10^{-3} = 0.05$

Sol.: Number of glucose molecules = $0.05\ N_A$

Number of carbon atoms = $6 \times 0.05\ N_A = 0.3\ N_A$

86. Answer (1)

Hint: $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$

Sol.: 1 mole $BaSO_4$ is produced by 1 mole $BaCl_2$

Now, 0.1 mole $BaSO_4$ = 0.1 mole $BaCl_2$

Mass of 0.1 mole $BaCl_2$ = 20.8 g

87. Answer (3)

Hint: An atom absorbs energy in form of discrete photon and energy of photon must be equal to difference of energy between two energy levels.

Sol.: Since 11 eV is not value of any ΔE, this photon will not be absorbed and electron will be present in ground state only.

88. Answer (1)

Hint: Electron can come back to ground state by single or multiple steps

Sol.: If only one H-atom is present in sample we will observe only one maximum line in visible spectrum.

89. Answer (1)

Hint: For single electron species

$2\pi r_n = n\lambda$

Sol.: $$2\pi \times 0.529 \times \frac{(3)^2}{3} = 3\lambda$$

$\lambda = 1.058\pi$ Å

90. Answer (4)

Hint: Maximum energetic photon will evolve if electron comes back to ground state from higher energy state.

Sol.: $4 \rightarrow 1$ will give most energetic photon.

[BOTANY]

91. Answer (3)

Hint: Nucleolus is the membrane less structure found in nucleoplasm.

Sol.: Nucleolus is a site for active ribosomal RNA synthesis. Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.

92. Answer (1)

Hint: Synthesis of proteins occur on the ribosomes attached to the ER.

Sol.: The correct sequence of translocation of protein molecule is RER → Transitional vesicles → Golgi apparatus → Secretory vesicles → Plasma membrane

93. Answer (4)

Hint: In some prokaryotes like cyanobacteria, there are membranous extensions containing pigments, into the cytoplasm called chromatophores.

Sol.: Several ribosomes may attach to a single mRNA and form a chain called polyribosome or polysome. The ribosomes of a polysome translate the mRNA into proteins.

Fimbriae are small bristle like fibres sprouting out of the cell and help them to attach to the host tissues.

94. Answer (3)

Hint: Centrosome is an organelle usually containing two cylindrical structures called centrioles.

Sol.: Cytoskeleton helps in the motility. Chloroplast has its own genetic material. Chromatin contains basic proteins such as histones.

95. Answer (2)

Hint: Chloroplast and mitochondria are semi-autonomous cell organelles.

Sol.: Lysosome is surrounded by single membrane.

96. Answer (3)

Hint: Basal body is the structure at the base of the cilium or flagellum that organises the axoneme formation.

Sol.: The core of cilia or flagella is called axoneme, which usually has nine doublets of peripheral microtubules and a pair of centrally located microtubule.

Basal body of cilia or flagella has 9+0 arrangement of microtubules.

97. Answer (2)

Hint: Vacuole is an organelle bound by a single membrane called tonoplast and it contains water, sap, excretory products and other materials useful for the cell.

Sol.: Mitochondria - Helps in oxidative phosphorylation

Chloroplast - Imparts green colour due to presence of chlorophyll

98. Answer (2)

Hint: The lipids are arranged within the membrane with the polar heads towards the outer side and the hydrophobic tail towards the inner part.

Sol.: In the membrane, the non-polar tail of saturated hydrocarbon is protected from aqueous solution and hence they are projected inside in the lipid bilayer.

99. Answer (2)

Hint: Plasmid is an extra chromosomal DNA in prokaryotic organism.

Sol.: In plant cell, both chloroplast and mitochondria have their own genetic material.

100. Answer (3)

Hint: Mesosomes are special membranous structure which is formed by extensions of plasma membrane.

Sol.: Mesosome helps in the formation of cell wall and DNA replication.

101. Answer (4)

Hint: A layer of glycocalyx is responsible for giving the bacteria a sticky character.

Sol.: The capsule is responsible for giving gummy and sticky character to the cell.

102. Answer (2)

Hint: The sequence of events by which a cell duplicates its genome, synthesises the other constituents of the cell and eventually divides into two daughter cells is termed cell cycle.

Sol.: All the processes, *i.e.*, cell division, DNA replication, and cell growth, have to take place in a coordinated way to ensure correct division and formation of progeny cells containing intact genomes.

103. Answer (4)

Hint: The final stage of meiotic prophase I is diakinesis.

Sol.: Diakinesis is marked by terminalisation of chiasmata.

104. Answer (1)

Hint: Mitochondria is the powerhouse of the cell.

Sol.: In mitochondria, cristae help in increasing the surface area for ATP synthesis. Its matrix contains 70S ribosomes, single circular DNA and ribosomes.

105. Answer (4)

Hint: Organelles involved in endomembrane system perform coordinated function.

Sol.: Peroxisomes are not included in endomembrane system.

106. Answer (3)

Hint: Carotenoids are fat-soluble pigments.

Sol.: Water-soluble pigment in plant cell vacuole is anthocyanin.

107. Answer (2)

Hint: Integral or transmembrane protein are embedded within the hydrophobic core of the phospholipids bilayer.

Sol.: Detergents are required for the extraction of integral membrane proteins. Chloroplast is semi-autonomous cell organelle. PPLO is not an autotrophic organism.

108. Answer (2)

Hint: Telocentric chromosome appear I shaped during the anaphase stage of the cell division.

Sol.: The type of chromosome in which the centromere is present very close to one end of the chromosome and they appear J shaped during anaphase is acrocentric.

109. Answer (1)

Hint: Bivalents are formed during the zygotene stage, when two homologous chromosomes undergo synapsis.

Sol.: The bivalent has four chromatids of the two homologous chromosomes.

110. Answer (2)

Hint: Metaphase II of meiosis has half the number of chromosomes as compared to the metphase I.

Sol.: In metaphase I bivalents are arranged on the equatorial plate. While in metaphase II, univalents are arranged on the equatorial plate.

111. Answer (3)

Sol.: Cell cycle is divided into two basic phases, interphase and M phase.

112. Answer (1)

Hint: Robert Brown identified the nucleus in the plant cell.

Sol.: Matthias Schleiden concluded that the plants are composed of different kinds of cells.

Rudolf Virchow proposed that every cell arises from a pre-existing cell.

113. Answer (3)

Hint: The first stage of interphase is G_1 phase, which is immediately begin after the completion of cytokinesis but before the onset of the DNA synthesis.

Sol.: Although the chromosome number remains the same in both G_1 and G_2 phases, but the amount of DNA doubles after the S phase.

114. Answer (1)

Hint: Centrioles are not universally essential for mitosis in all eukaryotes.

Sol.: In higher plants, centrioles are absent. Despite this, spindle fibres are assembled. These spindle fibres function normally in chromosome alignment and segregation.

115. Answer (2)

Hint: Cell P could be an animal cell and cell Q is a plant cell.

Sol.: In plant cells' cytokinesis occurs by cell plate formation through the fusion of Golgi-derived vesicles. Cell wall maintains the shape of the cell.

116. Answer (3)

Hint: DNA replication occurs only once during the process of meiosis

Sol.: Each daughter cell after meiosis I contained 46 chromosomes and 2C DNA.

117. Answer (3)

Hint: Mitosis is also known as the equational division.

Sol.: Genetic variability in the daughter cells is the result of meiosis.

118. Answer (3)

Hint: Each centrosome radiates out microtubule called asters and two asters together with spindle fibres form mitotic apparatus.

Sol.: Centromere is not a part of mitotic apparatus. It is the primary constriction of the chromosome. Cilia and flagella are originated from basal body.

119. Answer (1)

Hint: During the metaphase stage, nuclear envelope and nucleolus is no longer visible and chromosomes attain maximum degree of condensation.

Sol.: Metaphase stage is followed by the anaphase stage. During anaphase I, separation of homologous chromosomes takes place.

120. Answer (3)

Hint: In multicellular organisms, growth is considered in terms of an increase in cell number rather than merely an increase in cytoplasmic content.

Sol.: M phase of the cell cycle involves both nuclear division and cytoplasmic division. Both haploid and diploid cells in plants can show mitotic division.

121. Answer (3)

Hint: The transition phase between telophase I and prophase II is interkinesis.

Sol.: DNA replication does not take place during interkinesis

122. Answer (3)

Hint: Centrosomes radiate out microtubules and form a star shaped structure.

Sol.: The microtubules radiate out of the centrosomes are called asters.

123. Answer (1)

Hint: Golgi apparatus and semi-autonomous cell organelles divide in the G_2 phase of the cell cycle.

Sol.: The cells during G_1 phase are metabolically active and continuously grow but do not replicate their DNA.

124. Answer (4)

Hint: Splitting of centromere takes place during the third phase of the karyokinesis.

Sol.: The centromere of each chromosome splits during anaphase of mitosis and anaphase II of meiosis.

125. Answer (2)

Hint: Meiosis is a specialized type of cell division that reduces the chromosome number by half.

Sol.: Meiosis consists of one round of DNA replication followed by two successive nuclear divisions.

126. Answer (1)

Hint: Glycocalyx is the outermost layer of cell envelope in bacteria.

Sol.: Glycocalyx comprises of coating of mucous or polysaccharide macromolecule which protects the cells and also helps in adhesion.

127. Answer (2)

Hint: Middle lamella helps in joining of the adjacent cells in a plant tissue by a thin, sticky amorphous layer of cementing material.

Sol.: The cell wall and middle lamella are traversed by plasmodesmata which connects the cytoplasm of neighbouring cells.

128. Answer (3)

Hint: Centrosome is not surrounded by any membrane and they are not found in prokaryotes.

Sol.: Centrosome helps in the formation of spindle fibres.

129. Answer (4)

Sol.: The cells in the G_0 phase remains metabolically active but do not proliferate unless they are called on to do so.

130. Answer (1)

Hint: Karyokinesis is division of nucleus.

Sol.: The final stage of karyokinesis is telophase.

131. Answer (3)

Hint: Recombination of non-sister chromatids of homologous chromosome takes place in pachytene stage.

Sol.: Chiasmata appears during the diplotene stage of meiosis I. During diakinesis, terminalisation of chiasmata will occur.

132. Answer (1)

Hint: Contractile vacuoles are involved in osmoregulation in freshwater organism.

Sol.: Contractile vacuole perform excretion in *Amoeba*.

133. Answer (2)

Hint: 70S type of ribosomes are found in the semi-autonomous cell organelles.

Sol.: The two subunits of 70S ribosomes are 50S and 30S.

134. Answer (3)

Hint: White Blood cells are amoeboid.

Sol.: Human RBCs are round and biconcave. Nerve cells are branched and long.

135. Answer (4)

Hint: The flattened membranous sacs of chloroplast are thylakoids.

Sol.: Thylakoid contains chlorophyll pigments.

[ZOOLOGY]

136. Answer (3)

Hint: Secretion of endocrine gland

Sol.: Exocrine glands secrete mucus, saliva, earwax, oil, milk, digestive enzymes, *etc*. These products are released through ducts or tubes. In contrast, endocrine glands do not have ducts. Their products called hormones are secreted directly into the fluid bathing the gland.

137. Answer (3)

Hint: Function of communication junction

Sol.: Three types of cell junctions are found in the epithelium and other tissues. These are called tight, adhering and gap junctions. Tight junctions help to stop substances from leaking across a tissue. Adhering junctions perform cementing to keep neighbouring cells together. Gap junctions facilitate cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules.

138. Answer (1)

Hint: Location of ciliated columnar epithelium

Sol.: The columnar epithelium is composed of a single layer of tall and slender cells. Their nuclei are located at the base. If the free surface, bear cilia, then they are called ciliated epithelium. They are mainly present in the inner surface of hollow organs like oviduct, where fertilisation takes place.

139. Answer (2)

Hint: Exclude the type of cell junction present in cardiac muscles

Sol.: Intestinal epithelial cells have tight junctions in the apical part of the adjacent cells. These junctions seal the intercellular space between the adjacent cells, preventing bacteria, toxins, and undigested molecules from leaking into the underlying tissue and bloodstream. This barrier function is critical for maintaining intestinal integrity. Gap junctions are meant for communication.

140. Answer (3)

Hint: X = Proteins

Sol.: Water constitutes 70-90% of the total cellular mass. Proteins constitute 10-15% of the total cellular mass. Trypsin, antibody and collagen are proteinaceous in nature. Inulin is a homopolysaccharide.

141. Answer (1)

Hint: Monomeric unit of cellulose is glucose.

Sol.: In a polysaccharide chain (say cellulose), the right end is called the reducing end and the left end is called the non-reducing end. In a protein chain, the first amino acid is called N-terminal amino acid and the last amino acid is called the C-terminal amino acid.

142. Answer (4)

Hint: Possesses two nitrogen atoms.

Sol.: In a DNA strand, adenine and guanine of one strand compulsorily base pairs with thymine and cytosine, respectively, on the other strand [*i.e.,* A = T and G ≡ C]

In a RNA strand, instead of thymine, uracil is present.

143. Answer (2)

Hint: Polymer of glucose

Sol.:

Pigments	Carotenoids, Anthocyanins, *etc.*
Alkaloids	Morphine, Codeine, *etc.*
Terpenoides	Monoterpenes, Diterpenes, *etc.*
Essential oils	Lemon grass oil, *etc.*
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs	Vinblastin, curcumin, *etc.*
Polymeric substances	Rubber, gums, cellulose

144. Answer (2)

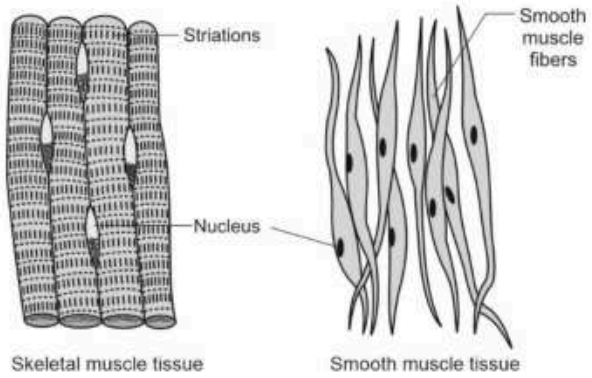
Hint: Areolar tissue belongs to the category of tissue to which adipose tissue also belongs to.

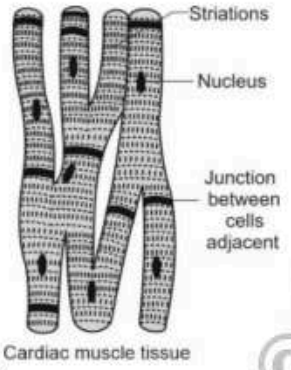
Sol.: In the dense regular connective tissue, the collagen fibres are present in rows between many parallel bundles of fibres. Tendons, which attach skeletal muscles to bones and ligaments which attach one bone to another are examples of this tissue. Areolar tissue is a loose connective tissue, where fibres are loosely arranged.

145. Answer (4)

Hint: Feature exclusive to cardiac muscles

Sol.:





146. Answer (1)

Hint: Areolar tissue

Sol.: Loose connective tissue has cells and fibres loosely arranged in a semi-fluid ground substance, for example, areolar tissue present beneath the skin. Often it serves as a support framework for epithelium. It contains fibroblasts, macrophages and mast cells. Muscles play an active role in all the movements of the body.

147. Answer (3)

Hint: Adipose tissue contains adipocytes.

Sol.: Adipose tissue is a good insulator and can therefore, reduce heat loss through the skin.

Neural tissue exerts the greatest control over the body's responsiveness to changing conditions.

Epithelial tissue has its own nerve supply, but it is avascular. Muscles maintain the positions of the various parts of the body.

148. Answer (4)

Hint: True for bones

Sol.: The intercellular material of cartilage is solid and pliable and resists compression. Cells of this tissue (Chondrocytes) are enclosed in small cavities within the matrix secreted by them. The bone marrow in some bones is the site of production of blood cells.

149. Answer (1)

Hint: Ash analysis gives elemental analysis of inorganic compounds.

Sol.: One weighs a small amount of living tissue and dry it. All the water, evaporates. The remaining material gives dry weight. If the tissue is fully burnt, all the carbon compounds are oxidised to gaseous form (CO_2, water vapour) and are removed, what is remaining is called ash. This ash contains inorganic elements like calcium, magnesium, *etc*.

Element	% Weight	
	Earth's crust	Human body
Hydrogen (H)	0.14	9.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	negligible

150. Answer (2)

Hint: Location of bone

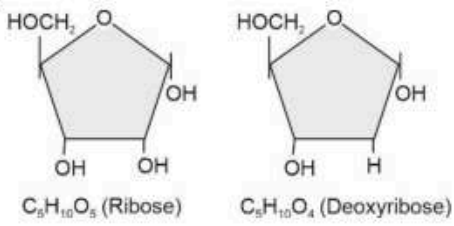
Sol.: Bones have a hard and non-pliable ground substance rich in calcium salts and collagen fibres which give bone its strength. Cartilage is present in the tip of nose, outer ear joints, between adjacent bones of the vertebral column.

Limb bones such as the long bones of the legs, serve weight-bearing functions.

151. Answer (1)

Hint: RNA stands for ribonucleic acid

Sol.:



152. Answer (3)

Hint: Free enzyme is released.

Sol.: Enzymes work by forming a transient enzyme-substrate complex, the transition state, thereby lowering the activation energy. At lower activation energy threshold, more substrate molecules have sufficient kinetic energy to undergo the reaction.

Enzymes are regenerated after each reaction cycle and do not change the equilibrium of the reaction.

153. Answer (3)

Hint: One of them is a nucleotide

Sol.: Cellulose is a linear homopolymer of β-D glucose units. Cytidylic acid is a nucleotide, in which the glycosidic bond connects the nitrogen base (Cytosine, at N1 of the pyrimidine ring) to the C1' of the pentose sugar. Thus, both cellulose and cytidylic acid contain glycosidic bond. Cytidylic acid is present in acid-soluble pool while cellulose is found in acid-insoluble fraction.

154. Answer (2)

Hint: Contains one glycerol molecule

Sol.: Chitin is found in exoskeleton of arthropods. It is a homopolysaccharide.

Triglyceride is composed of one glycerol and three fatty acid molecules. Triglyceride contains ester bonds.

GLUT-4 enables glucose transport in the cells and it is a protein.

Cholesterol is a derived lipid and it does not contain fatty acids but is included in the lipids as it has fat-like properties.

155. Answer (4)

Hint: Observed in myoglobin

Sol.: Lipids are also found in the retentate (acid-insoluble fraction) but are not polymeric. Ribozymes are RNA molecules. RNA contains phosphodiester bonds between nucleotides, not peptide bonds.

Only alanine has $-CH_3$ as its R-group; each amino acid has a unique R-group.

156. Answer (3)

Hint: Select the neutral amino acid.

Sol.: A particular property of amino acids is the ionisable nature of $-NH_2$ and $-COOH$ groups. Hence in solutions of different pH, the structure of amino acid changes.

Adenine – A purine derivative
Valine – Neutral amino acid
Vinblastin – Secondary metabolite
Cytosine – Pyrimidine nitrogenous base

157. Answer (4)

Hint: True for compound epithelium

Sol.: Compound epithelium covers the dry surface of the skin, the moist surface of buccal cavity, pharynx, inner lining of ducts of salivary glands and of pancreatic ducts. Compound epithelium is made of more than one layer (multi-layered) of cells and thus has a limited role in secretion and absorption. Their main function is to provide protection against chemical and mechanical stresses.

158. Answer (1)

Hint: PCT of nephron is lined by brush-bordered cuboidal epithelium.

Sol.: PCT is lined by simple cuboidal brush-border epithelium which increases the surface area for reabsorption.

Small intestine is lined by brush bordered columnar epithelium.

The columnar epithelium is composed of a single layer of tall and slender cells.

159. Answer (2)

Hint: Show excitability.

Sol.: Neurons are excitable cells but they do not divide. Muscle fibres contract (shorten) in response to stimulation, then relax (lengthen) and return to their uncontracted state in a coordinated fashion.

Blood contains erythrocytes, leucocytes and thrombocytes.

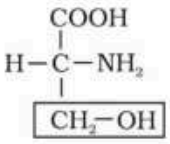
Some of the columnar or cuboidal cells get specialised for secretion and are called glandular epithelium.

160. Answer (2)

Hint: Lecithin possesses 2 fatty acid molecules

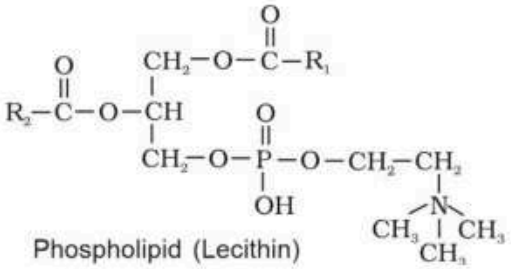


Uracil (Pyrimidine)



Serine

Sol.:



Phospholipid (Lecithin)

161. Answer (1)

Hint: Similar function performed by starch

Sol.: Glycogen is a storage homopolysaccharide. It is present as a store house of energy in animal tissues. It is present in liver as well as skeletal muscles.

162. Answer (4)

Hint: Can also be categorised as a drug.

Sol.:

Pigments	Carotenoids, Anthocyanins, *etc.*
Alkaloids	Morphine, Codeine, *etc.*
Terpenoides	Monoterpenes, Diterpenes, *etc.*
Essential oils	Lemon grass oil, *etc.*
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs	Vinblastin, curcumin, *etc.*
Polymeric substances	Rubber, gums, cellulose

163. Answer (2)

Hint: Follow the formula $C_nH_{(2n-2d)}O_2$

Sol.: For a fatty acid with 'n' carbons and 'd' double bonds, the formula is $C_nH_{(2n-2d)}O_2$. For arachidonic acid, n = 20, d = 4, thus, $C_{20}H_{32}O_2$ is the formula of arachidonic acid.

164. Answer (1)

Hint: At high temperature, proteins get denatured.

Sol.: When an egg is boiled, the egg white coagulates as heat breaks the weak bonds holding its structure together, a process called denaturation. The newly unfolded proteins cross-link with one another to form a solid gel network.

165. Answer (4)

Hint: Haemoglobin is present in RBCs

Sol.:

RuBisCO	-	Most abundant protein in whole of the biosphere
Haemoglobin	-	Exhibits quaternary structure and aids in gaseous transport
Alpha helix	-	Secondary structure of protein
Carbonic anhydrase	-	Catalyses the formation of carbonic acid.

166. Answer (3)

Hint: True for neural tissue

Sol.: Connective tissue consists of two basic elements: extracellular matrix and cells. The extracellular matrix consists of protein fibres, ground substance and the material between the cells and the fibres. In contrast to epithelia, connective tissues do not usually occur on body surfaces.

167. Answer (1)

Hint: Identify a skeletal muscle

Sol.: Smooth muscle fibres are uninucleated and they usually do not get fatigued. Smooth muscles are found in the posterior part of oesophagus, stomach, intestine, lungs, urogenital tract, urinary bladder, iris of eye, *etc.* Biceps possesses skeletal muscle fibres which are multinucleated and get easily fatigued.

168. Answer (2)

Hint: Composed of a single layer of flattened cells

Sol.: The simple squamous epithelium is made of a single thin layer of flattened cells with irregular boundaries. They are found in the walls of blood vessels and air sacs of lungs and are involved in forming a diffusion boundary.

Transitional epithelium is present in the walls of urinary bladder.

169. Answer (4)

Hint: Multi-layered epithelium

Sol.: The primary function of compound epithelium is to provide protection against chemical and mechanical stresses.

Simple squamous epithelium aids in diffusion and filtration.

Transitional epithelium permits distension.

170. Answer (3)

Hint: Mediators of allergy

Sol.: Mast cells are irregularly ovoid cells and contain basophilic granules.

These granules contain

(a)	Histamine	-	Inflammatory substance produced during allergic reactions
(b)	Heparin	-	Natural anti-coagulant
(c)	Serotonin	-	Vasoconstrictor

- Macrophages are phagocytic in nature.

171. Answer (1)

Hint: Include secretion from goblet cells

Sol.: Unicellular glands consist of isolated glandular cells (*e.g.*, goblet cell of alimentary canal) and multicellular gland consists of cluster of cells.

Salivary gland, sweat gland, mammary gland and sebaceous gland are examples of multicellular gland.

172. Answer (4)

Hint: V_{max} remains constant in presence of competitive inhibitor.

Sol.: Competitive inhibitors have close structural similarity with the substrate. The inhibitor competes with the substrate for the substrate-binding site of the enzymes. Such competitive inhibitors are often used in the control of bacterial pathogens. With the increase in substrate concentration, the velocity of the enzymatic reaction rises at first. The reaction ultimately reaches a maximum velocity (V_{max}) which is not exceeded by any further rise in concentration of the substrate.

173. Answer (2)

Hint: One of them is a basic amino acid.

Sol.: Lysine is a basic amino acid and it has an additional amino group.

Valine is a neutral amino acid and it has equal number of amino and carboxylic groups.

Tyrosine is an aromatic amino acid.

174. Answer (1)

Hint: Both are polysaccharides

Sol.: Starch and glycogen both are polymers of glucose and serve as a storage form in plants and animals, respectively.

Glycogen as well as starch possess a branched structure.

Glycogen gives red colour with iodine and starch gives blue colour with iodine.

175. Answer (2)

Hint: Exclude a polysaccharide

Sol.: Carboxypeptidase, antibody, receptor and trypsin are proteins while chitin is a homopolysaccharide found in the exoskeleton of arthropods.

176. Answer (4)

Hint: Two glucose units are joined by dehydration

Sol.: For joining of 2 units of glucose $\rightarrow$ 1 glycosidic bond is formed by removal of one water molecule.

For n units in a linear chain, the number of bonds is (n – 1)

Accordingly, n – 1 = 230

n = 231

177. Answer (3)

Hint: Exclude cartilage cells

Sol.: Fibroblasts are the most common cells in loose connective tissue and are responsible for synthesising and secreting the fibres and ground substance of the extracellular matrix. Plasma cells secrete antibodies.

178. Answer (4)

Hint: Exhibits 3-dimensional view

Sol.: α-helix and β-pleated sheet are two types of secondary structures. In α-helix, there is interaction between every 4th amino acid by the formation of intramolecular hydrogen bond.

Active site of enzymes exhibits tertiary structure.

179. Answer (3)

Hint: Transferases

Sol.: Transferases are the enzymes that catalyse the transfer of specific groups, other than hydrogen, from one substrate to another.

180. Answer (4)

Hint: Not true for prosthetic group.

Sol.: In peroxidase and catalase, which catalyse the breakdown of hydrogen peroxide to water and oxygen, haem is the prosthetic group and is the part of active site of enzyme. Co-enzyme NADP contains vitamin niacin.

❑ ❑ ❑

All India Aakash Test Series for NEET - 2028

TEST - I (Code-J)

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Test Date : 01/11/2026

ANSWERS

1.	(3)	37.	(3)	73.	(1)	109.	(1)	145.	(1)
2.	(1)	38.	(2)	74.	(2)	110.	(2)	146.	(3)
3.	(2)	39.	(1)	75.	(4)	111.	(3)	147.	(4)
4.	(1)	40.	(2)	76.	(3)	112.	(1)	148.	(2)
5.	(1)	41.	(2)	77.	(4)	113.	(3)	149.	(1)
6.	(3)	42.	(4)	78.	(3)	114.	(1)	150.	(3)
7.	(4)	43.	(1)	79.	(3)	115.	(2)	151.	(4)
8.	(2)	44.	(2)	80.	(3)	116.	(3)	152.	(1)
9.	(2)	45.	(3)	81.	(3)	117.	(3)	153.	(2)
10.	(2)	46.	(4)	82.	(4)	118.	(3)	154.	(4)
11.	(1)	47.	(1)	83.	(3)	119.	(1)	155.	(1)
12.	(3)	48.	(1)	84.	(2)	120.	(3)	156.	(2)
13.	(2)	49.	(3)	85.	(3)	121.	(3)	157.	(2)
14.	(1)	50.	(1)	86.	(1)	122.	(3)	158.	(1)
15.	(1)	51.	(4)	87.	(2)	123.	(1)	159.	(4)
16.	(2)	52.	(1)	88.	(4)	124.	(4)	160.	(3)
17.	(1)	53.	(4)	89.	(1)	125.	(2)	161.	(4)
18.	(1)	54.	(4)	90.	(1)	126.	(1)	162.	(2)
19.	(3)	55.	(2)	91.	(3)	127.	(2)	163.	(3)
20.	(3)	56.	(2)	92.	(1)	128.	(3)	164.	(3)
21.	(3)	57.	(3)	93.	(4)	129.	(4)	165.	(1)
22.	(3)	58.	(3)	94.	(3)	130.	(1)	166.	(2)
23.	(4)	59.	(4)	95.	(2)	131.	(3)	167.	(1)
24.	(2)	60.	(1)	96.	(3)	132.	(1)	168.	(4)
25.	(4)	61.	(1)	97.	(2)	133.	(2)	169.	(3)
26.	(3)	62.	(4)	98.	(2)	134.	(3)	170.	(1)
27.	(4)	63.	(2)	99.	(2)	135.	(4)	171.	(4)
28.	(4)	64.	(2)	100.	(3)	136.	(4)	172.	(2)
29.	(3)	65.	(2)	101.	(4)	137.	(3)	173.	(2)
30.	(2)	66.	(4)	102.	(2)	138.	(4)	174.	(4)
31.	(1)	67.	(1)	103.	(4)	139.	(3)	175.	(1)
32.	(1)	68.	(4)	104.	(1)	140.	(4)	176.	(3)
33.	(1)	69.	(2)	105.	(4)	141.	(2)	177.	(2)
34.	(2)	70.	(3)	106.	(3)	142.	(1)	178.	(1)
35.	(2)	71.	(3)	107.	(2)	143.	(2)	179.	(3)
36.	(4)	72.	(4)	108.	(2)	144.	(4)	180.	(3)

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HINTS & SOLUTIONS

[PHYSICS]

1. Answer (3)

Hint: Write G in terms of c, g and p as $[G] = [c^a g^b p^c]$ and now write dimension of both sides of equation and equate dimension of M, L and T, to find a, b and c.

Sol.: $[G] = [LT^{-1}]^a [LT^{-2}]^b \cdot [ML^{-1}T^{-2}]^c$

$\Rightarrow \left[\frac{MLT^{-2} \times L^2}{M \cdot M}\right] = [M^c L^{a+b-c} T^{-a-2b-2c}]$

$\Rightarrow [M^{-1}L^3T^{-2}] = [M^c L^{a+b-c} T^{-a-2b-2c}]$

$\Rightarrow c = -1,\ a + b - c = 3$ and $-a - 2b - 2c = -2$

from these equations we have

$c = -1,\ a + b = 2$ ...(i)

and $a + 2b = 4$ ...(ii)

After solving (i) and (ii), we get

$c = -1,\ a = 0$ and $b = 2$

so, $[G] = [c^0 g^2 p^{-1}]$

2. Answer (1)

Hint and Sol.: Only multiplication or division is possible for quantities having different dimensions.

3. Answer (2)

Hint: Errors are very small, so, differentiate the equation and find $\frac{dy}{y}$ to find fractional error in y

Sol.: $y = k\tan x$

$\Rightarrow dy = k\sec^2 x dx$

$\Rightarrow \frac{dy}{y} = \frac{\sec^2 x dx}{\tan x} = \frac{dx}{\cos^2 x \times \frac{\sin x}{\cos x}}$

$= \frac{dx}{\sin x \cos x} = \frac{2dx}{\sin 2x}$

4. Answer (1)

Hint and Sol.: Here digits to be dropped are 6 and 7, so preceding digit is increased by one.

5. Answer (1)

Hint and Sol.: Using rules for counting significant figures, 0.01 has one significant figure and 0.50 has two significant figure and (R) explains (A).

6. Answer (3)

Hint: There will be error of $\frac{1}{5}$ second in measured time $t = 15$ second and here $t = nT$

Sol.: $t = nT$

$\Rightarrow \Delta t = n\Delta T$

$\Rightarrow \frac{\Delta t}{t} = \frac{\Delta T}{T}$

$\therefore \frac{\Delta t}{t} = \frac{\frac{1}{5}}{15} = \frac{1}{75}$

Percentage error in $(T) = \frac{100}{75} = \frac{4}{3} = 1.3\%$

7. Answer (4)

Hint: Focus on definition of physical quantity and write expression

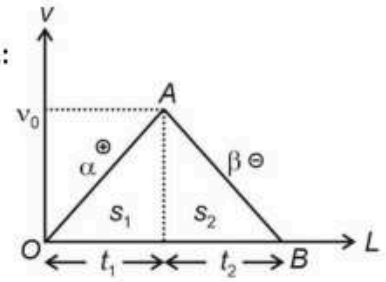
$s = \frac{E}{A \times t}$

Sol.: $[s] = \frac{[ML^2T^{-2}]}{[L^2 \times T]} = [ML^0T^{-3}]$

8. Answer (2)

Hint: Draw v-t graph and use $v = u + at$ as well as $s = \frac{1}{2}at^2$

Sol.:



For motion O to A:

$v_0 = \alpha t_1 \Rightarrow t_1 = \frac{v_0}{\alpha}$ ...(i)

For motion A to B:

$0 = v_0 - \beta t_2$

$\Rightarrow t_2 = \frac{v_0}{\beta}$ ...(ii)

$\Rightarrow \frac{t_1}{t_2} = \frac{\beta}{\alpha}$

Now area of *v-t* graph gives displacement

$\Rightarrow$ *O* to *A*:

$s_1 = \frac{1}{2} v_0 t_1$ ...(iii)

and *A* to *B*:

$s_2 = \frac{1}{2} v_0 t_2$ ...(iv)

$\Rightarrow \frac{s_1}{s_2} = \frac{t_1}{t_2}$

9. Answer (2)

Hint: Use $v = \frac{dx}{dt}$ and $a = \frac{dv}{dt}$

Sol.: $x = \frac{1}{2}t^2 - \frac{1}{3}t^3$

$v = \frac{dx}{dt} = \frac{d}{dt}\left[\frac{1}{2}t^2 - \frac{1}{3}t^3\right]$

$= \frac{1}{2} \times 2t - \frac{1}{3} \times 3t^2$

$v = t - t^2$

and $a = \frac{dv}{dt} = \frac{d}{dt}(t - t^2)$

$a = 1 - 2t$

here $a = 0$

$\Rightarrow 1 - 2t = 0$

$t = \frac{1}{2}$ second

10. Answer (2)

Hint: Use, $v = u + at$

Sol.: $-3u = u - gt$

$\Rightarrow t = \frac{4u}{g}$

11. Answer (1)

Hint: Learn it, as a standard case or use $h = ut - \frac{1}{2}gt^2$ and solve for its roots t_1 and t_2

Sol.: Here $h = ut - \frac{1}{2}gt^2$

It has two values (roots) t_1 and t_2 for same *h*.

so, converting this equation in standard quadratic form

$2h = 2ut - gt^2$

$\Rightarrow gt^2 - 2ut + 2h = 0$

Now using condition for products of roots for $ax^2 + bx + c = 0$

$\Rightarrow \alpha\beta = \frac{c}{a}$

$\Rightarrow t_1 t_2 = \frac{2h}{g}$

$\Rightarrow h = \frac{1}{2} g t_1 t_2$

12. Answer (3)

Hint and Sol.: $a = \frac{dv}{dt}$ *i.e.*, slope of *v-t* graph gives acceleration.

For first half-time slope is negative and constant while for second half-time slope is positive and constant, so option (3) is correct.

13. Answer (2)

Hint and Sol.: *s* always denotes displacement covered while *u* and *a* may be positive, or negative or zero.

14. Answer (1)

Hint: Time taken to cross each other

$= \frac{\text{Sum of total length of trains}}{\text{relative speed of trains}}$

Sol.: Crossing time $= \frac{100 + 60}{|v_A| + |v_B|}$

$4 = \frac{160}{v + 3v} = \frac{160}{4v}$

$\Rightarrow v = \frac{160}{16} = 10$ m/s

so $v_A = 10$ m/s and $v_B = 30$ m/s

15. Answer (1)

Hint and Sol.: Distance covered is always increasing for a moving body.

Therefore speed, $\frac{ds}{dt} \rightarrow$ positive

16. Answer (2)

Hint: For uniform acceleration

$v_{av} = \frac{u + v}{2}$ and

$s = v_{av} \times$ time

Sol.: $v = \dfrac{0+144}{2} = 72$ km/h

$= \dfrac{72\times5}{18} = 20$ m/s

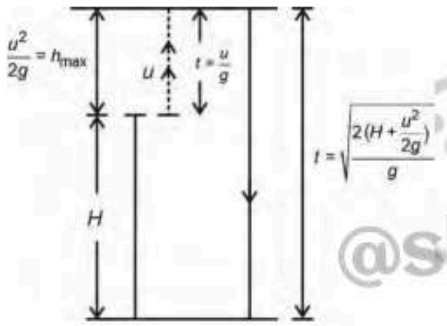
Now $s = v_{av} \times t = 20\times20 = 400$ m

17. Answer (1)

Hint: Maximum height attained by stone (dropped from B_1) is least, so total time of journey for stone dropped from B_1 is least.

Sol.: Total journey time for a stone is given by

$$T = \frac{u}{g} + \sqrt{\frac{2\left(H + \frac{u^2}{2g}\right)}{g}}$$



From above expression of T, it is clear that

If u is higher, then T will be higher

18. Answer (1)

Hint: Slope of position(x)-time(t) graph gives velocity.

Sol.: From graph it is clear that at 6 points slope is zero *i.e.*, $v = 0$, so particle comes to rest 6 times.

19. Answer (3)

Hint: Use, $h = ut - \dfrac{1}{2}gt^2$

Sol.: Using $h = ut - \dfrac{1}{2}gt^2$

According to question

$h_1 = h_2$

$\Rightarrow ut - \dfrac{1}{2}gt^2 = u(t-5) - \dfrac{1}{2}g(t-5)^2$

$\Rightarrow ut - 5t^2 = u(t-5) - 5(t-5)^2$

$\Rightarrow -5t^2 = -5u - 5(t-5)^2$

$\Rightarrow -t^2 = -u - (t-5)^2$

$\Rightarrow t^2 - (t-5)^2 = 40 \quad [\because u = 40$ m/s$]$

$\Rightarrow t^2 - t^2 + 10t - 25 = 40$

$10t = 65 \Rightarrow t = 6.5$ second

Now, $h = ut - \dfrac{1}{2}gt^2$

$h = 40\times6.5 - \dfrac{1}{2}10\times(6.5)^2$

$= 260 - 5 \times 6.5 \times 6.5$

$= 260 - 211.25 = 48.75$ metre

20. Answer (3)

Hint: Instantaneous acceleration, $a = \dfrac{dv}{dt}$

Sol.: $A = \dfrac{dv}{dt} \rightarrow dv = A \cdot dt$

$\Rightarrow \int dv = \int A dt$

$\Rightarrow \int_u^v dv = \int_0^t at \cdot dt = \dfrac{at^2}{2}$

$\Rightarrow v - u = \dfrac{at^2}{2} \Rightarrow v = u + \dfrac{at^2}{2}$

21. Answer (3)

Hint: $v_{PQ} = v_P - v_Q$ and slope of v-t graph gives acceleration

Sol.: $v_P = a_P t$ and

$v_Q = a_Q t$

$\therefore \quad v_{PQ} = v_P - v_Q = (a_P - a_Q)t$

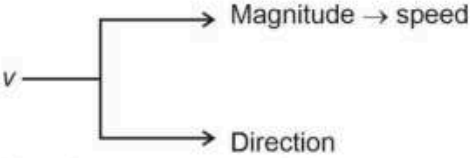
here $a_P - a_Q$ is positive constant

$\Rightarrow v_{PQ} = Kt$

i.e., relative velocity will increase linearly with time.

22. Answer (3)

Hint and Sol.: $a = \dfrac{dv}{dt}$; and



From here we can conclude

If velocity is changing, we cannot surely say whether speed is changing or not as well as whether direction is changing or not.

23. Answer (4)

Hint and Sol.: Acceleration, $a = \frac{dv}{dt}$

from this, (i) if $v \rightarrow$ constant

then $a = 0$ at all time

(ii) If v is continuously varying then $\frac{dv}{dt} \neq 0$, even $v = 0$ at any instant (in case of reversal point of motion)

24. Answer (2)

Hint: Here acceleration and velocity are in opposite direction so particle will return back on same path. We have to check reversal instant (when $v = 0$) is in between given time interval or outside the given interval. Then we decide whether, distance is equal to displacement or not; if not, then distance = actual path length

Sol.: At reversal point $v = 0$

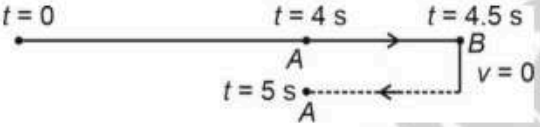
then time at which it occurs

$v = u + at$

$\Rightarrow 0 = 9 - 2t \qquad \Rightarrow t = 4.5$ second

so this is in between $t = 4$ s to $t = 5$ s

(i.e., fifth second)



here required distance = $|AB| + |BA|$

$= 2|BA|$ ($\because$ acceleration $\rightarrow$ constant)

$= 2 \times \frac{1}{2} \times 2 \times \left(\frac{1}{2}\right)^2$

$= 2 \times \frac{1}{4} = \frac{1}{2} = 0.5$ m

25. Answer (4)

Hint: Use $v^2 = u^2 + 2as$

Sol.: $v^2 = u^2 + 2as$

$0 = u^2 - 2as$

$s = \frac{u^2}{2a}$

here $a \rightarrow$ constant

$\therefore \quad s \propto u^2$

when u is made n times, then

s will become n^2 times.

26. Answer (3)

Hint and Sol.:

Acceleration, $a = \frac{dv}{dt} \Rightarrow \int dv = \int a dt$

$\Rightarrow \Delta v = \int a dt$ = area under acceleration vs time graph

27. Answer (4)

Hint: Use the concept of relative velocity and equations of uniformly accelerated motion.

Sol.: Stopping distance $= \frac{u^2}{2a} = \frac{(4)^2}{2 \times 1.2} = 6.66$ m < 10 m

$\Rightarrow$ He will never be able to catch bus.

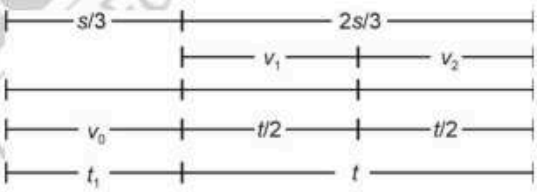
28. Answer (4)

Hint and Sol.: For coming to rest, acceleration and velocity must be in opposite direction. So, angle should be 180° (π radian)

29. Answer (3)

Hint: Average speed $= \frac{\text{Total distance covered}}{\text{Time taken}}$

Sol.:



here, $t_1 = \frac{s}{3v_0}$, $v_1\frac{t}{2} + \frac{v_2}{2}t = 2\frac{s}{3}$

$\Rightarrow t = \frac{4s}{3(v_1 + v_2)}$

Now, Average speed $= \frac{s}{t_1 + t}$

$= \frac{s}{\frac{s}{3v_0} + \frac{4s}{3(v_1 + v_2)}}$

$= \frac{1}{\left[(v_1 + v_2) + 4v_0\right] / 3v_0(v_1 + v_2)}$

$= \frac{3v_0(v_1 + v_2)}{(v_1 + v_2) + 4v_0}$

30. Answer (2)

Hint: Use $s_{n^{th}} = u + \frac{a}{2}[2n-1]$

Sol.: $\frac{s_{3rd}}{s_{5th}} = \frac{S_1}{S_2} = \frac{\frac{a}{2}[2\times 3-1]}{\frac{a}{2}[2\times 5-1]} = \frac{5}{9}$

31. Answer (1)

Hint and Sol.: In case of uniform motion, speed and velocity are constant. Therefore, all statements A, B and C are correct

32. Answer (1)

Hint: Use velocity $v = \frac{dx}{dt}$

and acceleration $a = \frac{dv}{dt}$

Sol.: $x = (2t+5)^2$

$\Rightarrow x = 4t^2 + 20t + 25$

$\Rightarrow v = \frac{dx}{dt} = 8t + 20$

$\Rightarrow a = \frac{dv}{dt} = 8$ = constant

33. Answer (1)

Hint and Sol.:

(A-Q) $v = u - gt = -gt + u$

Graph will be straight line with negative slope

(B-R) $|v| = |u - gt|$

Graph will be always positive and straight line

(C-P) $h = ut - \frac{1}{2}gt^2$ parabola with downward opening.

34. Answer (2)

Hint: $LC = \frac{\text{Pitch}(x)}{\text{Number of divisions on circular scale } (n)}$

Sol.: $LC = \frac{x}{n}$

Initially $0.01 = \frac{x}{n}$

finally, $LC' = \frac{\left(x + \frac{75}{100}x\right)}{n - \frac{n\times 50}{100}}$

$LC' = \frac{\frac{175}{100}x}{\frac{n}{2}}$

$LC' = \frac{350}{100}\times\frac{x}{n} = 3.5\times 0.01 = 0.035$ mm

35. Answer (2)

Hint: In multiplication/division, final result will have significant figures as equal to any data which has least significant figure.

Sol.: Here least significant figures are three so simply by observation we can mark option (2) because x = 1330 has 3 significant figure.

Note: You can also do mathematical operation, and then finally round-off the result upto 3 significant figure.

36. Answer (4)

Hint: Area under this v-t graph gives distance covered.

Sol.: Area bounded in t = 30.5 second

$= \frac{1}{2}(400+200)\times 2 + 400\times(30.5-2)$

= 300 × 2 + 400 × 28.5

= 600 + 11400

= 12000 m = 12 km

37. Answer (3)

Hint: Use principle of dimensional homogeneity i.e., each term has same dimensions.

Sol.: $[A]$ = [L]

$[B]$ = [L]

$[C] = [LT^{-2}]$

and $[D]$ = [L]

$\therefore \left[\frac{ABC}{D}\right] = \left[\frac{L\times L\times LT^{-2}}{L}\right] = [L^2T^{-2}]$

38. Answer (2)

Hint: Density, $\rho = \frac{m}{V} = \frac{m}{\pi r^2 l}$

Sol.: $\Rightarrow \left(\frac{\Delta\rho}{\rho}\right)_{\text{maximum}} = \left(\frac{\Delta m}{m} + \frac{2\Delta r}{r} + \frac{\Delta l}{l}\right)$

$\left(\frac{\Delta\rho}{\rho}\right)_{max} = \frac{0.003}{0.60} + \frac{2\times 0.01}{0.50} + \frac{0.05}{10.00}$

$= \frac{0.1}{20} + \frac{2}{50} + \frac{5}{1000}$

$= \frac{5+40+5}{1000} = \frac{50}{1000} = \frac{5}{100}$

Maximum percentage error in density = 5%

39. Answer (1)

Hint and Sol.:

$$\text{Power} = \frac{\text{Work}}{\text{Time}} \rightarrow \frac{\text{Joule}}{\text{Second}} \rightarrow \text{Watt}$$

So, kilowatt $\rightarrow 10^3$ watt

40. Answer (2)

Hint and Sol.: $\text{Velocity} = \dfrac{\text{Displacement}}{\text{time}}$

$= \dfrac{\text{Length}}{\text{Time}}$

$\therefore$ Velocity is derived quantity

41. Answer (2)

Hint: Use principle of dimensional homogeneity and we know that exponent terms and their powers are dimensionless.

Sol.: $x = \dfrac{v_0}{\alpha} - \dfrac{v_0}{\alpha} e^{-\alpha t}$

$\Rightarrow$ αt is dimensionless

$[\alpha] = \left[\dfrac{1}{t}\right]$

$[\alpha] = [T^{-1}]$

and $[x] = \left[\dfrac{v_0}{\alpha}\right]$

$\Rightarrow [L] = \dfrac{[v_0]}{[T^{-1}]}$

$\Rightarrow [v_0] = [LT^{-1}]$

$\therefore \alpha^2 v_0^3 \rightarrow [T^{-2}] \times [LT^{-1}]^3$

$= [L^3T^{-5}] = [M^0L^3T^{-5}]$

42. Answer (4)

Hint: Check, if dimensions of LHS are equal to RHS, then equation is dimensionally correct otherwise incorrect.

Sol.: All first three options are standard results and they are dimensionally correct but option (4) is dimensionally incorrect.

since term (u) $\rightarrow$ speed $\rightarrow [LT^{-1}]$

and term $\left[\dfrac{1}{2}at^2\right] \rightarrow [LT^{-2}] \times [T^2] = [LT^0] = [L]$

and $[S] \rightarrow [L]$

So option (4) is dimensionally incorrect.

43. Answer (1)

Hint and Sol.: Using binomial expansion *i.e.*,

$(1+x)^n \approx 1 + nx$ if $|x| << 1$

$\therefore \left(1 + \dfrac{h}{R}\right)^{-2} \approx 1 - \dfrac{2h}{R}$

44. Answer (2)

Hint: Recall formula for surface area, $A = 6a^2$

Sol.: $A = 6a^2$ ...(1)

$\Rightarrow \dfrac{\Delta A}{A} = \dfrac{2\Delta a}{a}$

$= 2 \times 1\% = 2\%$

45. Answer (3)

Hint & Sol.: Least precise measurement may or may not be most accurate and vice-versa.

[CHEMISTRY]

46. Answer (4)

Hint: Maximum energetic photon will evolve if electron comes back to ground state from higher energy state.

Sol.: $4 \rightarrow 1$ will give most energetic photon.

47. Answer (1)

Hint: For single electron species

$2\pi r_n = n\lambda$

Sol.: $2\pi \times 0.529 \times \dfrac{(3)^2}{3} = 3\lambda$

$\lambda = 1.058\pi$ Å

48. Answer (1)

Hint: Electron can come back to ground state by single or multiple steps

Sol.: If only one H-atom is present in sample we will observe only one maximum line in visible spectrum.

49. Answer (3)

Hint: An atom absorbs energy in form of discrete photon and energy of photon must be equal to difference of energy between two energy levels.

Sol.: Since 11 eV is not value of any ΔE, this photon will not be absorbed and electron will be present in ground state only.

50. Answer (1)

Hint: $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$

Sol.: 1 mole $BaSO_4$ is produced by 1 mole $BaCl_2$

Now, 0.1 mole $BaSO_4$ = 0.1 mole $BaCl_2$

Mass of 0.1 mole $BaCl_2$ = 20.8 g

51. Answer (4)

Hint: Mole of glucose = $200 \times 0.25 \times 10^{-3} = 0.05$

Sol.: Number of glucose molecules = $0.05\ N_A$

Number of carbon atoms = $6 \times 0.05\ N_A = 0.3\ N_A$

52. Answer (1)

Hint: $\bar{\nu} = R_H \times Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

Sol.: $n_2 = 5, n_1 = 2$

$$\bar{\nu} = R_H \times (2)^2\left(\frac{1}{4} - \frac{1}{25}\right) = R_H\frac{21}{25}$$

53. Answer (4)

Hint: For single electron energy of orbital depends only on principal quantum number.

Sol.: H^- is two electron species

Hence 2nd excited state will be 2*p* subshell where value of *n* is 2.

54. Answer (4)

Hint: For single electron species

$$E_n = \frac{-13.6Z^2}{n^2}$$

Sol.: $$\frac{(E_3)_{Be^{3+}}}{(E_3)_{Li^{2+}}} = \frac{\frac{-13.6\times(4)^2}{(3)^2}}{\frac{-13.6\times(3)^2}{(3^2)}}$$

$$\frac{(E_3)_{Be^{3+}}}{x} = \frac{16}{9}$$

$$(E_3)_{Be^{3+}} = \frac{16x}{9}$$

55. Answer (2)

Hint: KE = E_{photon} – Work function

Sol.: KE = $6.6 \times 10^{-34} \times 10^{15} - 3.44 \times 10^{-19}$

$= 3.16 \times 10^{-19}$ J

56. Answer (2)

Hint: For a black-body radiation, wavelength of intensity maxima decreases with increasing temperature.

Sol.: Since $\lambda_3 < \lambda_2 < \lambda_1$

hence $T_3 > T_2 > T_1$

57. Answer (3)

Hint: Electron has maximum e/m ratio.

Sol.: Microwave has higher frequency than radio wave.

58. Answer (3)

Hint: The element with atomic number 24 is chromium.

Sol.: Cr: $3d^5 4s^1$

$\therefore$ 6 unpaired electrons are present.

59. Answer (4)

Hint: $Cu \Rightarrow [Ar]3d^{10}4s^1$

Sol.: Number of electron with $m_l = 0$ are 13

60. Answer (1)

Hint: Number of orbital in a shell = n^2

Sol.: Number of orbital in 4th shell = $4^2 = 16$

As atomic number increases energy of orbital decreases.

For monoelectronic species energy depends on principal quantum number only.

Splitting of spectral line in magnetic field is Zeeman effect.

61. Answer (1)

Hint: Number of spherical node = n – l – 1

Sol.: For 5*p* orbital

Number of spherical mole = 5 – 1 – 1 = 3

d_{xy} orbital has two nodal planes.

62. Answer (4)

Hint: 'n' is a positive integer which is always greater than 'm'

Sol.: m, varies from '+l' to '–l'

s can be only $\frac{+1}{2}$ or $\frac{-1}{2}$

63. Answer (2)

Hint: de-Broglie wavelength

$$\lambda = \frac{h}{\sqrt{2mKE}}$$

Sol.: $$\lambda = \frac{6.6\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\times28\times10^{-23}}}$$

$$= \frac{6.6\times10^{-34}}{7.13\times10^{-27}} = 0.93 \times 10^{-7}\ m$$

64. Answer (2)

Hint: For microscopic species

$$\Delta x\Delta p \geq \frac{h}{4\pi}$$

Sol.: $\Delta x\Delta v = \frac{h}{4\pi m}$

$$(\Delta x)^2 = \frac{h}{4\pi m}$$

$$\Delta x = \left(\frac{h}{4\pi m}\right)^{1/2}$$

$$\sqrt{\frac{h}{4\pi m}} \times \Delta p = \frac{h}{4\pi}$$

$$\Delta p = \sqrt{\frac{hm}{4\pi}}$$

65. Answer (2)

Hint: For Mn atom last electron will be filled in 3*d* orbital.

Sol.:

Mg $\Rightarrow$ n = 3, l = 0, m = 0, $s = +\frac{1}{2}$

K $\Rightarrow$ n = 4, l = 0, m = 0, $s = +\frac{1}{2}$

Br $\Rightarrow$ n = 4, l = 1, m = –1, $s = +\frac{1}{2}$

Mn $\Rightarrow$ n = 3, l = 2, m = +1, $s = +\frac{1}{2}$

66. Answer (4)

Hint: Avg atomic mass $\frac{\sum A_i X_i}{100}$

$A_i \Rightarrow$ mass number of ith isotope

$X_i \Rightarrow$ percentage abundance of

Sol.: % abundance of heavier isotope = x

% abundance of lighter isotope = 100 – x

$$\frac{x \times 22 + (100 - x)20}{100} = 20.3$$

22x + 2000 –20x = 2030

2x = 30

x = 15%

67. Answer (1)

Hint: Isobars have same mass number but different atomic numbers

Sol.: $A_1 = Z_A + n_A$

100 = 48 + n_A

n_A = 52

$A_2 = Z_B + n_B$

100 = 52 +n_B

n_B = 48

68. Answer (4)

Hint: According to JJ Thomson atom is uniform sphere of positive charge in which negative charge is embedded into it in such a manner as to give the most stable electrostatic arrangement.

Sol.: Presence of nucleus in an atom is confirmed by Rutherford's α-particle experiment.

$$\frac{r_{nucleus}}{r_{atom}} = 10^5$$

$$\frac{m_{proton}}{m_{electron}} = \frac{1.67 \times 10^{-27}\ kg}{9.109 \times 10^{-31}\ kg} = 1837$$

69. Answer (2)

Hint: For neutralisation number of gram equivalents of acid must be equal to number of gram equivalents of base.

Sol.: $(n_{eq})_{acid} = 100 \times 0.1 + 100 \times 0.5 = 60$

$(n_{eq})_{NaOH} = 60$

0.2 × V = 60

V = 300 mL

70. Answer (3)

Hint: $E_{metal\ oxide} = E_m + E_0$

Sol.: $\frac{E_m}{E_0} = \frac{60}{40}$

$$E_m = \frac{3}{2} \times 8 = 12$$

$E_{metal\ oxide}$ = 12 + 8 = 20

71. Answer (3)

Hint: $°F = \left(\frac{9}{5}\right) \times °C + 32$

Sol.: For 25°C,

$$F = \frac{9}{5} \times 25 + 32 = 77°F$$

For 40°C,

$$F = \frac{9}{5} \times 40 + 32 = 104°F$$

For –60°C,

$$F = \frac{9}{5} \times (-60) + 32 = -76°F$$

@shivamsp

For 100°C,

$$F = \frac{9}{5} \times 100 + 32 = 200°F$$

72. Answer (4)

Hint: S is limiting reagent for given reaction

Sol.:

$$Mg(s) + S(s) \rightarrow MgS(s)$$

$$\frac{4}{24} \quad \frac{4}{32}$$

$$\frac{1}{6} \text{ mol} \quad \frac{1}{8} \text{ mol}$$

mole of MgS = $\frac{1}{8}$

mass of MgS $= \frac{1}{8} \times 56 = 7$ g

73. Answer (1)

Hint: $C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$

Sol.: Number of mole of $C_2H_6 = \frac{3}{30} = 0.1$

1 mole C_2H_6 requires 3.5 mol O_2

0.1 mole C_2H_6 requires 0.35 mol O_2 = 11.2 g

Volume of CO_2 obtained at STP = 0.1 × 2 × 22.4 = 4.48 L

74. Answer (2)

Hint: % H = 100 – (%C + %O)

= 100 – (54.54 + 36.37)

= 9.09%

Sol.:

	C	H	O
Mass% :	54.54	9.09	36.37
Mole ratio :	54.54/12	9.09/1	36.37/16
	4.54	9.09	2.27
Simplest ratio :	2	4	1

Empirical formula C_2H_4O

Empirical formula mass = 44 u

75. Answer (4)

Hint: For minimum molecular mass, given molecule must have 1, Mg atom

Sol.: $\%Mg = \frac{1 \times 24}{M} \times 100$

$0.01 = \frac{24}{M} \times 100$

M = 2.4 × 10^5 u

76. Answer (3)

Hint: Mole of 0.15 g $CO_3^{2-} = \frac{0.15}{60}$ = 2.5 × 10^{-3}

Sol.: Number of ions = 2.5 × 10^{-3} × 6.02 × 10^{23}

= 1.5 × 10^{21} ions

Number of O atoms = 3 × 1.5 × 10^{21} = 4.5 × 10^{21}

mass of carbon = 0.0025 × 12 = 0.03 g

77. Answer (4)

Hint: For mixing

$$[H^+] = \frac{N_1V_1 + N_2V_2 + N_3V_3}{V_1 + V_2 + V_3}$$

Sol.: $[H^+] = \frac{200 \times 0.4 + 300 \times 0.1 + 500 \times 0.4}{200 + 300 + 500}$

$= \frac{80 + 30 + 200}{1000}$ = 0.31 M

78. Answer (3)

Hint: Volume of one mole gas at STP is 22.4 L

Sol.:

$$d = \frac{\text{mass}}{\text{volume}}$$

$= \frac{28}{22.4}$ = 1.25 g/L

79. Answer (3)

Hint: 6.00 has three significant figure while 600 has only one.

Sol.: Mass percentage of oxygen in glucose $= \frac{16 \times 6 \times 100}{180} = 53.3\%$.

80. Answer (3)

Hint: Mole of HNO_3 = 700 × 3 × 10^{-3} = 2.1

Sol.: Mass of HNO_3 = 2.1 × 63 = 132.3 g

70 g HNO_3 is present in 100 g of solution

132.3 g HNO_3 is present in $\frac{100 \times 132.3}{70} = 189$ g

81. Answer (3)

Hint: According to Dalton atomic theory the matter is composed of extremely small indivisible particles.

Sol.: Atoms of different element combine with each other in a fixed whole number ratio to form compounds

82. Answer (4)

Hint: 1000 g of solvent contains 180 g of urea

Sol.: Mass of solution = (1000 + 180) g = 1180 g

1.18 kg of solution contains 1000 g of water

1.8 kg of solution contains $\frac{1000\times1.8}{1.18}$ g of water

Mass of water = 1.5 kg

83. Answer (3)

Hint: Empirical formula represents simplest ratio of number of atoms of constituents present in one molecule.

Sol.: For benzene molecular formula is C_6H_6 and empirical formula is CH and for acetylene molecular formula is C_2H_2 and empirical formula is CH.

84. Answer (2)

Hint: $N_2 + 3H_2 \rightarrow 2NH_3$

H_2 will be limiting reagent

Sol.:

	N_2	+ $3H_2$	$\rightarrow 2NH_3$
t = 0	14 g	2 g	
	0.5 mol	1 mol	0
t = t	0.5 – 0.33	0	0.66 mol

mass of NH_3 formed $= \frac{2}{3}\times17 = 11.33$ g

volume of NH_3 produced at STP $= \frac{2}{3}\times22.4$

= 14.93 L

85. Answer (3)

Hint: Only acidic gas gets absorbed in basic KOH pellets

Sol.:

	$HCOOH \xrightarrow{\text{conc. } H_2SO_4}$	$H_2O(l)$ +	$CO(g)$
t = 0	0.1 mol	0	0
t = t	0	0.1	0.1

	$H_2C_2O_4 \xrightarrow{\text{conc. } H_2SO_4}$	$H_2O(l)$ +	$CO(g)$ +	$CO_2(g)$
t = 0	0.1 mol	0	0	0
t = t	0	0.1	0.1	0.1

Since CO_2 will get absorbed in KOH, remaining gaseous product is CO gas only

mole of CO gas = 0.2

mass of CO gas = 0.2 × 28

= 5.6 g

86. Answer (1)

Hint: On heating, water of hydration will be removed.

Sol.: $Na_2SO_4\cdot xH_2O \xrightarrow{\Delta} Na_2SO_4 + xH_2O$

Mole(s) of anhydrous $NaSO_4 = \frac{\text{mole of } H_2O}{x}$

$\frac{44.1}{142} = \frac{55.9}{18\times x}$

x = 10

87. Answer (2)

Hint: $2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$

$2KCl + H_2SO_4 \rightarrow K_2SO_4 + 2HCl$

Sol.: 117 g NaCl gives 142 g Na_2SO_4

hence x g NaCl gives $\frac{142}{117}\times$'x' g Na_2SO_4

(1.8062 – x) g KCl will give

$\frac{174}{149}(1.8062 - x)$ g K_2SO_4

$\frac{142x}{117} + \frac{174(1.8062-x)}{149} = 2.1568$

21158x + 36770.62 – 20358x = 37599.49

800x = 828.87

x = 1.03 g

1.8062 – x = 0.7701 g

$\%NaCl = \frac{1.03}{1.8062}\times100$

= 57.36%

%KCl = 42.64%

88. Answer (4)

Hint: Number of molecule = mole N_A

Sol.: Initial molecules $= \frac{2.24\times10^{-3}}{22.4}\times N_A$

$= 10^{-4}\ N_A$

Removed molecules $= \frac{2.1\times10^{-3}}{28}\times N_A$

$= 0.75 \times 10^{-4}\ N_A$

Left molecule = $10^{-4}\ N_A - 0.75 \times 10^{-4}\ N_A$

$= 0.25 \times 10^{-4}\ N_A$

$= 2.5 \times 10^{-5} \times 6.02 \times 10^{23}$

$= 1.5 \times 10^{19}$

89. Answer (1)

Hint: Kilogram is SI unit of mass.

Sol.: Mole ⇒ Amount of substance

Candela ⇒ Luminous intensity

Ampere ⇒ Electric current

90. Answer (1)

Hint: Homogeneous mixture has uniform composition throughout the mixture

Sol.: Aqueous NaCl and air are homogeneous mixture.

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[BOTANY]

91. Answer (3)

Hint: Nucleolus is the membrane less structure found in nucleoplasm.

Sol.: Nucleolus is a site for active ribosomal RNA synthesis. Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.

92. Answer (1)

Hint: Synthesis of proteins occur on the ribosomes attached to the ER.

Sol.: The correct sequence of translocation of protein molecule is RER → Transitional vesicles → Golgi apparatus → Secretory vesicles → Plasma membrane

93. Answer (4)

Hint: In some prokaryotes like cyanobacteria, there are membranous extensions containing pigments, into the cytoplasm called chromatophores.

Sol.: Several ribosomes may attach to a single mRNA and form a chain called polyribosome or polysome. The ribosomes of a polysome translate the mRNA into proteins.

Fimbriae are small bristle like fibres sprouting out of the cell and help them to attach to the host tissues.

94. Answer (3)

Hint: Centrosome is an organelle usually containing two cylindrical structures called centrioles.

Sol.: Cytoskeleton helps in the motility. Chloroplast has its own genetic material. Chromatin contains basic proteins such as histones.

95. Answer (2)

Hint: Chloroplast and mitochondria are semi-autonomous cell organelles.

Sol.: Lysosome is surrounded by single membrane.

96. Answer (3)

Hint: Basal body is the structure at the base of the cilium or flagellum that organises the axoneme formation.

Sol.: The core of cilia or flagella is called axoneme, which usually has nine doublets of peripheral microtubules and a pair of centrally located microtubule.

Basal body of cilia or flagella has 9+0 arrangement of microtubules.

97. Answer (2)

Hint: Vacuole is an organelle bound by a single membrane called tonoplast and it contains water, sap, excretory products and other materials useful for the cell.

Sol.: Mitochondria - Helps in oxidative phosphorylation

Chloroplast - Imparts green colour due to presence of chlorophyll

98. Answer (2)

Hint: The lipids are arranged within the membrane with the polar heads towards the outer side and the hydrophobic tail towards the inner part.

Sol.: In the membrane, the non-polar tail of saturated hydrocarbon is protected from aqueous solution and hence they are projected inside in the lipid bilayer.

99. Answer (2)

Hint: Plasmid is an extra chromosomal DNA in prokaryotic organism.

Sol.: In plant cell, both chloroplast and mitochondria have their own genetic material.

100. Answer (3)

Hint: Mesosomes are special membranous structure which is formed by extensions of plasma membrane.

Sol.: Mesosome helps in the formation of cell wall and DNA replication.

101. Answer (4)

Hint: A layer of glycocalyx is responsible for giving the bacteria a sticky character.

Sol.: The capsule is responsible for giving gummy and sticky character to the cell.

102. Answer (2)

Hint: The sequence of events by which a cell duplicates its genome, synthesises the other constituents of the cell and eventually divides into two daughter cells is termed cell cycle.

Sol.: All the processes, *i.e.*, cell division, DNA replication, and cell growth, have to take place in a coordinated way to ensure correct division and formation of progeny cells containing intact genomes.

103. Answer (4)

Hint: The final stage of meiotic prophase I is diakinesis.

Sol.: Diakinesis is marked by terminalisation of chiasmata.

104. Answer (1)

Hint: Mitochondria is the powerhouse of the cell.

Sol.: In mitochondria, cristae help in increasing the surface area for ATP synthesis. Its matrix contains 70S ribosomes, single circular DNA and ribosomes.

105. Answer (4)

Hint: Organelles involved in endomembrane system perform coordinated function.

Sol.: Peroxisomes are not included in endomembrane system.

106. Answer (3)

Hint: Carotenoids are fat-soluble pigments.

Sol.: Water-soluble pigment in plant cell vacuole is anthocyanin.

107. Answer (2)

Hint: Integral or transmembrane protein are embedded within the hydrophobic core of the phospholipids bilayer.

Sol.: Detergents are required for the extraction of integral membrane proteins. Chloroplast is semi-autonomous cell organelle. PPLO is not an autotrophic organism.

108. Answer (2)

Hint: Telocentric chromosome appear I shaped during the anaphase stage of the cell division.

Sol.: The type of chromosome in which the centromere is present very close to one end of the chromosome and they appear J shaped during anaphase is acrocentric.

109. Answer (1)

Hint: Bivalents are formed during the zygotene stage, when two homologous chromosomes undergo synapsis.

Sol.: The bivalent has four chromatids of the two homologous chromosomes.

110. Answer (2)

Hint: Metaphase II of meiosis has half the number of chromosomes as compared to the metphase I.

Sol.: In metaphase I bivalents are arranged on the equatorial plate. While in metaphase II, univalents are arranged on the equatorial plate.

111. Answer (3)

Sol.: Cell cycle is divided into two basic phases, interphase and M phase.

112. Answer (1)

Hint: Robert Brown identified the nucleus in the plant cell.

Sol.: Matthias Schleiden concluded that the plants are composed of different kinds of cells.

Rudolf Virchow proposed that every cell arises from a pre-existing cell.

113. Answer (3)

Hint: The first stage of interphase is G_1 phase, which is immediately begin after the completion of cytokinesis but before the onset of the DNA synthesis.

Sol.: Although the chromosome number remains the same in both G_1 and G_2 phases, but the amount of DNA doubles after the S phase.

114. Answer (1)

Hint: Centrioles are not universally essential for mitosis in all eukaryotes.

Sol.: In higher plants, centrioles are absent. Despite this, spindle fibres are assembled. These spindle fibres function normally in chromosome alignment and segregation.

115. Answer (2)

Hint: Cell P could be an animal cell and cell Q is a plant cell.

Sol.: In plant cells' cytokinesis occurs by cell plate formation through the fusion of Golgi-derived vesicles. Cell wall maintains the shape of the cell.

116. Answer (3)

Hint: DNA replication occurs only once during the process of meiosis

Sol.: Each daughter cell after meiosis I contained 46 chromosomes and 2C DNA.

117. Answer (3)

Hint: Mitosis is also known as the equational division.

Sol.: Genetic variability in the daughter cells is the result of meiosis.

118. Answer (3)

Hint: Each centrosome radiates out microtubule called asters and two asters together with spindle fibres form mitotic apparatus.

Sol.: Centromere is not a part of mitotic apparatus. It is the primary constriction of the chromosome. Cilia and flagella are originated from basal body.

119. Answer (1)

Hint: During the metaphase stage, nuclear envelope and nucleolus is no longer visible and chromosomes attain maximum degree of condensation.

Sol.: Metaphase stage is followed by the anaphase stage. During anaphase I, separation of homologous chromosomes takes place.

120. Answer (3)

Hint: In multicellular organisms, growth is considered in terms of an increase in cell number rather than merely an increase in cytoplasmic content.

Sol.: M phase of the cell cycle involves both nuclear division and cytoplasmic division. Both haploid and diploid cells in plants can show mitotic division.

121. Answer (3)

Hint: The transition phase between telophase I and prophase II is interkinesis.

Sol.: DNA replication does not take place during interkinesis

122. Answer (3)

Hint: Centrosomes radiate out microtubules and form a star shaped structure.

Sol.: The microtubules radiate out of the centrosomes are called asters.

123. Answer (1)

Hint: Golgi apparatus and semi-autonomous cell organelles divide in the G_2 phase of the cell cycle.

Sol.: The cells during G_1 phase are metabolically active and continuously grow but do not replicate their DNA.

124. Answer (4)

Hint: Splitting of centromere takes place during the third phase of the karyokinesis.

Sol.: The centromere of each chromosome splits during anaphase of mitosis and anaphase II of meiosis.

125. Answer (2)

Hint: Meiosis is a specialized type of cell division that reduces the chromosome number by half.

Sol.: Meiosis consists of one round of DNA replication followed by two successive nuclear divisions.

126. Answer (1)

Hint: Glycocalyx is the outermost layer of cell envelope in bacteria.

Sol.: Glycocalyx comprises of coating of mucous or polysaccharide macromolecule which protects the cells and also helps in adhesion.

127. Answer (2)

Hint: Middle lamella helps in joining of the adjacent cells in a plant tissue by a thin, sticky amorphous layer of cementing material.

Sol.: The cell wall and middle lamella are traversed by plasmodesmata which connects the cytoplasm of neighbouring cells.

128. Answer (3)

Hint: Centrosome is not surrounded by any membrane and they are not found in prokaryotes.

Sol.: Centrosome helps in the formation of spindle fibres.

129. Answer (4)

Sol.: The cells in the G_0 phase remains metabolically active but do not proliferate unless they are called on to do so.

130. Answer (1)

Hint: Karyokinesis is division of nucleus.

Sol.: The final stage of karyokinesis is telophase.

131. Answer (3)

Hint: Recombination of non-sister chromatids of homologous chromosome takes place in pachytene stage.

Sol.: Chiasmata appears during the diplotene stage of meiosis I. During diakinesis, terminalisation of chiasmata will occur.

132. Answer (1)

Hint: Contractile vacuoles are involved in osmoregulation in freshwater organism.

Sol.: Contractile vacuole perform excretion in *Amoeba*.

133. Answer (2)

Hint: 70S type of ribosomes are found in the semi-autonomous cell organelles.

Sol.: The two subunits of 70S ribosomes are 50S and 30S.

134. Answer (3)

Hint: White Blood cells are amoeboid.

Sol.: Human RBCs are round and biconcave. Nerve cells are branched and long.

135. Answer (4)

Hint: The flattened membranous sacs of chloroplast are thylakoids.

Sol.: Thylakoid contains chlorophyll pigments.

[ZOOLOGY]

136. Answer (4)

Hint: Not true for prosthetic group.

Sol.: In peroxidase and catalase, which catalyse the breakdown of hydrogen peroxide to water and oxygen, haem is the prosthetic group and is the part of active site of enzyme. Co-enzyme NADP contains vitamin niacin.

137. Answer (3)

Hint: Transferases

Sol.: Transferases are the enzymes that catalyse the transfer of specific groups, other than hydrogen, from one substrate to another.

138. Answer (4)

Hint: Exhibits 3-dimensional view

Sol.: α-helix and β-pleated sheet are two types of secondary structures. In α-helix, there is interaction between every 4th amino acid by the formation of intramolecular hydrogen bond.

Active site of enzymes exhibits tertiary structure.

139. Answer (3)

Hint: Exclude cartilage cells

Sol.: Fibroblasts are the most common cells in loose connective tissue and are responsible for synthesising and secreting the fibres and ground substance of the extracellular matrix. Plasma cells secrete antibodies.

140. Answer (4)

Hint: Two glucose units are joined by dehydration

Sol.: For joining of 2 units of glucose $\rightarrow$ 1 glycosidic bond is formed by removal of one water molecule.

For n units in a linear chain, the number of bonds is (n – 1)

Accordingly, n – 1 = 230

n = 231

141. Answer (2)

Hint: Exclude a polysaccharide

Sol.: Carboxypeptidase, antibody, receptor and trypsin are proteins while chitin is a homopolysaccharide found in the exoskeleton of arthropods.

142. Answer (1)

Hint: Both are polysaccharides

Sol.: Starch and glycogen both are polymers of glucose and serve as a storage form in plants and animals, respectively.

Glycogen as well as starch possess a branched structure.

Glycogen gives red colour with iodine and starch gives blue colour with iodine.

143. Answer (2)

Hint: One of them is a basic amino acid.

Sol.: Lysine is a basic amino acid and it has an additional amino group.

Valine is a neutral amino acid and it has equal number of amino and carboxylic groups.

Tyrosine is an aromatic amino acid.

144. Answer (4)

Hint: V_{max} remains constant in presence of competitive inhibitor.

Sol.: Competitive inhibitors have close structural similarity with the substrate. The inhibitor competes with the substrate for the substrate-binding site of the enzymes. Such competitive inhibitors are often used in the control of bacterial pathogens. With the increase in substrate concentration, the velocity of the enzymatic reaction rises at first. The reaction ultimately reaches a maximum velocity (V_{max}) which is not exceeded by any further rise in concentration of the substrate.

145. Answer (1)

Hint: Include secretion from goblet cells

Sol.: Unicellular glands consist of isolated glandular cells (*e.g.*, goblet cell of alimentary canal) and multicellular gland consists of cluster of cells.

Salivary gland, sweat gland, mammary gland and sebaceous gland are examples of multicellular gland.

146. Answer (3)

Hint: Mediators of allergy

Sol.: Mast cells are irregularly ovoid cells and contain basophilic granules.

These granules contain

(a)	Histamine	-	Inflammatory substance produced during allergic reactions
(b)	Heparin	-	Natural anti-coagulant
(c)	Serotonin	-	Vasoconstrictor

- Macrophages are phagocytic in nature.

147. Answer (4)

Hint: Multi-layered epithelium

Sol.: The primary function of compound epithelium is to provide protection against chemical and mechanical stresses.

Simple squamous epithelium aids in diffusion and filtration.

Transitional epithelium permits distension.

148. Answer (2)

Hint: Composed of a single layer of flattened cells

Sol.: The simple squamous epithelium is made of a single thin layer of flattened cells with irregular boundaries. They are found in the walls of blood vessels and air sacs of lungs and are involved in forming a diffusion boundary.

Transitional epithelium is present in the walls of urinary bladder.

149. Answer (1)

Hint: Identify a skeletal muscle

Sol.: Smooth muscle fibres are uninucleated and they usually do not get fatigued. Smooth muscles are found in the posterior part of oesophagus, stomach, intestine, lungs, urogenital tract, urinary bladder, iris of eye, *etc*. Biceps possesses skeletal muscle fibres which are multinucleated and get easily fatigued.

150. Answer (3)

Hint: True for neural tissue

Sol.: Connective tissue consists of two basic elements: extracellular matrix and cells. The extracellular matrix consists of protein fibres, ground substance and the material between the cells and the fibres. In contrast to epithelia, connective tissues do not usually occur on body surfaces.

151. Answer (4)

Hint: Haemoglobin is present in RBCs

Sol.:

RuBisCO	-	Most abundant protein in whole of the biosphere
Haemoglobin	-	Exhibits quaternary structure and aids in gaseous transport
Alpha helix	-	Secondary structure of protein
Carbonic anhydrase	-	Catalyses the formation of carbonic acid.

152. Answer (1)

Hint: At high temperature, proteins get denatured.

Sol.: When an egg is boiled, the egg white coagulates as heat breaks the weak bonds holding its structure together, a process called denaturation. The newly unfolded proteins cross-link with one another to form a solid gel network.

153. Answer (2)

Hint: Follow the formula $C_nH_{(2n-2d)}O_2$

Sol.: For a fatty acid with 'n' carbons and 'd' double bonds, the formula is $C_nH_{(2n-2d)}O_2$. For arachidonic acid, n = 20, d = 4, thus, $C_{20}H_{32}O_2$ is the formula of arachidonic acid.

154. Answer (4)

Hint: Can also be categorised as a drug.

Sol.:

Pigments	Carotenoids, Anthocyanins, *etc.*
Alkaloids	Morphine, Codeine, *etc.*
Terpenoides	Monoterpenes, Diterpenes, *etc.*
Essential oils	Lemon grass oil, *etc.*
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs	Vinblastin, curcumin, *etc.*
Polymeric substances	Rubber, gums, cellulose

155. Answer (1)

Hint: Similar function performed by starch

Sol.: Glycogen is a storage homopolysaccharide. It is present as a store house of energy in animal tissues. It is present in liver as well as skeletal muscles.

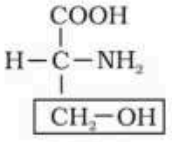
156. Answer (2)

Hint: Lecithin possesses 2 fatty acid molecules

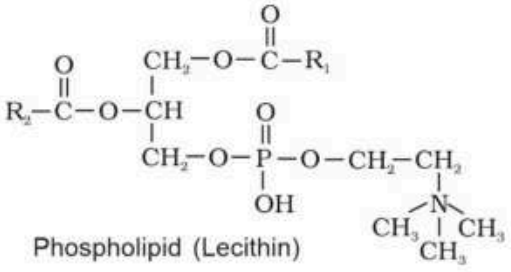
Sol.:



Uracil (Pyrimidine)



Serine



Phospholipid (Lecithin)

157. Answer (2)

Hint: Show excitability.

Sol.: Neurons are excitable cells but they do not divide. Muscle fibres contract (shorten) in response to stimulation, then relax (lengthen) and return to their uncontracted state in a coordinated fashion.

Blood contains erythrocytes, leucocytes and thrombocytes.

Some of the columnar or cuboidal cells get specialised for secretion and are called glandular epithelium.

158. Answer (1)

Hint: PCT of nephron is lined by brush-bordered cuboidal epithelium.

Sol.: PCT is lined by simple cuboidal brush-border epithelium which increases the surface area for reabsorption.

Small intestine is lined by brush bordered columnar epithelium.

The columnar epithelium is composed of a single layer of tall and slender cells.

159. Answer (4)

Hint: True for compound epithelium

Sol.: Compound epithelium covers the dry surface of the skin, the moist surface of buccal cavity, pharynx, inner lining of ducts of salivary glands and of pancreatic ducts. Compound epithelium is made of more than one layer (multi-layered) of cells and thus has a limited role in secretion and absorption. Their main function is to provide protection against chemical and mechanical stresses.

160. Answer (3)

Hint: Select the neutral amino acid.

Sol.: A particular property of amino acids is the ionisable nature of $-NH_2$ and $-COOH$ groups. Hence in solutions of different pH, the structure of amino acid changes.

Adenine – A purine derivative

Valine – Neutral amino acid

Vinblastin – Secondary metabolite

Cytosine – Pyrimidine nitrogenous base

161. Answer (4)

Hint: Observed in myoglobin

Sol.: Lipids are also found in the retentate (acid-insoluble fraction) but are not polymeric. Ribozymes are RNA molecules. RNA contains phosphodiester bonds between nucleotides, not peptide bonds.

Only alanine has $-CH_3$ as its R-group; each amino acid has a unique R-group.

162. Answer (2)

Hint: Contains one glycerol molecule

Sol.: Chitin is found in exoskeleton of arthropods. It is a homopolysaccharide.

Triglyceride is composed of one glycerol and three fatty acid molecules. Triglyceride contains ester bonds.

GLUT-4 enables glucose transport in the cells and it is a protein.

Cholesterol is a derived lipid and it does not contain fatty acids but is included in the lipids as it has fat-like properties.

163. Answer (3)

Hint: One of them is a nucleotide

Sol.: Cellulose is a linear homopolymer of β-D glucose units. Cytidylic acid is a nucleotide, in which the glycosidic bond connects the nitrogen base (Cytosine, at N1 of the pyrimidine ring) to the C1' of the pentose sugar. Thus, both cellulose and cytidylic acid contain glycosidic bond. Cytidylic acid is present in acid-soluble pool while cellulose is found in acid-insoluble fraction.

164. Answer (3)

Hint: Free enzyme is released.

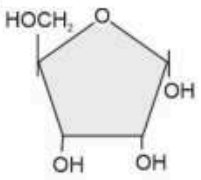
Sol.: Enzymes work by forming a transient enzyme-substrate complex, the transition state, thereby lowering the activation energy. At lower activation energy threshold, more substrate molecules have sufficient kinetic energy to undergo the reaction.

Enzymes are regenerated after each reaction cycle and do not change the equilibrium of the reaction.

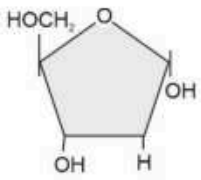
165. Answer (1)

Hint: RNA stands for ribonucleic acid

Sol.:



$C_5H_{10}O_5$ (Ribose)



$C_5H_{10}O_4$ (Deoxyribose)

166. Answer (2)

Hint: Location of bone

Sol.: Bones have a hard and non-pliable ground substance rich in calcium salts and collagen fibres which give bone its strength. Cartilage is present in the tip of nose, outer ear joints, between adjacent bones of the vertebral column.

Limb bones such as the long bones of the legs, serve weight-bearing functions.

167. Answer (1)

Hint: Ash analysis gives elemental analysis of inorganic compounds.

Sol.: One weighs a small amount of living tissue and dry it. All the water, evaporates. The remaining material gives dry weight. If the tissue is fully burnt, all the carbon compounds are oxidised to gaseous form (CO_2, water vapour) and are removed, what is remaining is called ash. This ash contains inorganic elements like calcium, magnesium, *etc*.

Element	% Weight	
	Earth's crust	Human body
Hydrogen (H)	0.14	9.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	negligible

168. Answer (4)

Hint: True for bones

Sol.: The intercellular material of cartilage is solid and pliable and resists compression. Cells of this tissue (Chondrocytes) are enclosed in small cavities within the matrix secreted by them. The bone marrow in some bones is the site of production of blood cells.

169. Answer (3)

Hint: Adipose tissue contains adipocytes.

Sol.: Adipose tissue is a good insulator and can therefore, reduce heat loss through the skin.

Neural tissue exerts the greatest control over the body's responsiveness to changing conditions.

Epithelial tissue has its own nerve supply, but it is avascular. Muscles maintain the positions of the various parts of the body.

170. Answer (1)

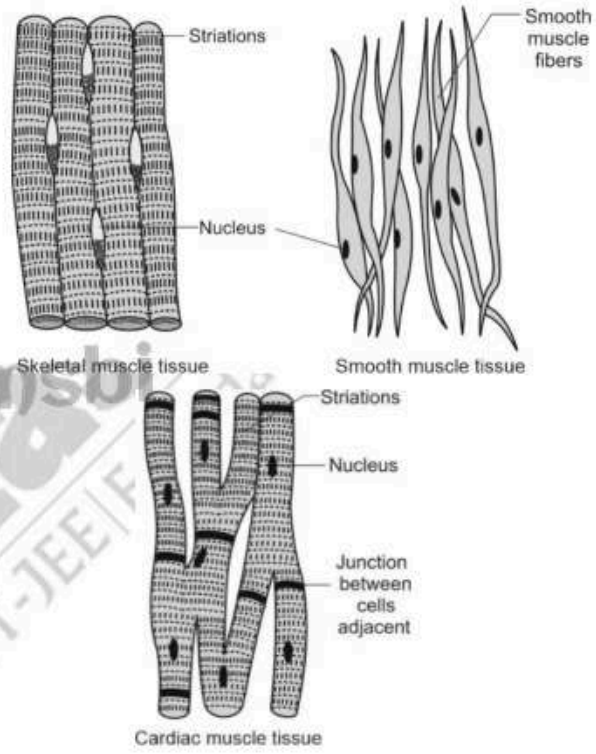
Hint: Areolar tissue

Sol.: Loose connective tissue has cells and fibres loosely arranged in a semi-fluid ground substance, for example, areolar tissue present beneath the skin. Often it serves as a support framework for epithelium. It contains fibroblasts, macrophages and mast cells. Muscles play an active role in all the movements of the body.

171. Answer (4)

Hint: Feature exclusive to cardiac muscles

Sol.:



172. Answer (2)

Hint: Areolar tissue belongs to the category of tissue to which adipose tissue also belongs to.

Sol.: In the dense regular connective tissue, the collagen fibres are present in rows between many parallel bundles of fibres. Tendons, which attach skeletal muscles to bones and ligaments which attach one bone to another are examples of this tissue. Areolar tissue is a loose connective tissue, where fibres are loosely arranged.

173. Answer (2)

Hint: Polymer of glucose

Sol.:

Pigments	Carotenoids, Anthocyanins, *etc.*
Alkaloids	Morphine, Codeine, *etc.*
Terpenoides	Monoterpenes, Diterpenes, *etc.*
Essential oils	Lemon grass oil, *etc.*
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs	Vinblastin, curcumin, *etc.*
Polymeric substances	Rubber, gums, cellulose

174. Answer (4)

Hint: Possesses two nitrogen atoms.

Sol.: In a DNA strand, adenine and guanine of one strand compulsorily base pairs with thymine and cytosine, respectively, on the other strand [*i.e.*, A = T and G ≡ C]

In a RNA strand, instead of thymine, uracil is present.

175. Answer (1)

Hint: Monomeric unit of cellulose is glucose.

Sol.: In a polysaccharide chain (say cellulose), the right end is called the reducing end and the left end is called the non-reducing end. In a protein chain, the first amino acid is called N-terminal amino acid and the last amino acid is called the C-terminal amino acid.

176. Answer (3)

Hint: X = Proteins

Sol.: Water constitutes 70-90% of the total cellular mass. Proteins constitute 10-15% of the total cellular mass. Trypsin, antibody and collagen are proteinaceous in nature. Inulin is a homopolysaccharide.

177. Answer (2)

Hint: Exclude the type of cell junction present in cardiac muscles

Sol.: Intestinal epithelial cells have tight junctions in the apical part of the adjacent cells. These junctions seal the intercellular space between the adjacent cells, preventing bacteria, toxins, and undigested molecules from leaking into the underlying tissue and bloodstream. This barrier function is critical for maintaining intestinal integrity. Gap junctions are meant for communication.

178. Answer (1)

Hint: Location of ciliated columnar epithelium

Sol.: The columnar epithelium is composed of a single layer of tall and slender cells. Their nuclei are located at the base. If the free surface, bear cilia, then they are called ciliated epithelium. They are mainly present in the inner surface of hollow organs like oviduct, where fertilisation takes place.

179. Answer (3)

Hint: Function of communication junction

Sol.: Three types of cell junctions are found in the epithelium and other tissues. These are called tight, adhering and gap junctions. Tight junctions help to stop substances from leaking across a tissue. Adhering junctions perform cementing to keep neighbouring cells together. Gap junctions facilitate cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules.

180. Answer (3)

Hint: Secretion of endocrine gland

Sol.: Exocrine glands secrete mucus, saliva, earwax, oil, milk, digestive enzymes, *etc.* These products are released through ducts or tubes. In contrast, endocrine glands do not have ducts. Their products called hormones are secreted directly into the fluid bathing the gland.

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